



7b. Large Language Models

Never Stand Still

Faculty of Engineering

COMP9444 7b

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The Rise of Language Models



Photo Credit: Image by Free-photos from Pixabay. <https://pixabay.com/photos/road-winding-street-bridge-1030789/>

Motivation

- A robot wrote this entire article. Are you scared yet, human?

<https://www.theguardian.com/commentisfree/2020/sep/08/robot-wrote-this-article-gpt-3>

- What It's Like To be a Computer: An Interview with GPT-3

https://www.youtube.com/watch?v=PqbB07n_uQ4

Natural Language Processing

- Enabling machines to process, represent, understand, and generate languages

In fact, the **Chinese** **NORP** market has the **three** **CARDINAL** most influential names of the retail and tech space – **Alibaba** **GPE**, **Baidu** **ORG**, and **Tencent** **PERSON** (collectively touted as **BAT** **ORG**), and is betting big in the global **AI** **GPE** in retail industry space. The **three** **CARDINAL** giants which are claimed to have a cut-throat competition with the **U.S.** **GPE** (in terms of resources and capital) are positioning themselves to become the ‘future **AI** **PERSON** platforms’. The trio is also expanding in other **Asian** **NORP** countries and investing heavily in the **U.S.** **GPE** based **AI** **GPE** startups to leverage the power of **AI** **GPE**. Backed by such powerful initiatives and presence of these conglomerates, the market in APAC AI is forecast to be the fastest-growing **one** **CARDINAL**, with an anticipated **CAGR** **PERSON** of **45%** **PERCENT** over **2018 - 2024** **DATE**.

To further elaborate on the geographical trends, **North America** **LOC** has procured **more than 50%** **PERCENT** of the global share in **2017** **DATE** and has been leading the regional landscape of **AI** **GPE** in the retail market. The **U.S.** **GPE** has a significant credit in the regional trends with **over 65%** **PERCENT** of investments (including M&As, private equity, and venture capital) in artificial intelligence technology. Additionally, the region is a huge hub for startups in tandem with the presence of tech titans, such as **Google** **ORG**, **IBM** **ORG**, and **Microsoft** **ORG**.

NLP Applications

➤ Text Classification

"I love this movie.
I've seen it many times
and it's still awesome."



"This movie is bad.
I don't like it at all.
It's terrible."



NLP Applications

➤ Machine Translation

Google Translate



Sign in

Text

Documents

DETECT LANGUAGE

ENGLISH

SPANISH

FRENCH



GERMAN

ENGLISH

SPANISH



I love teaching humans and machines



Ich liebe es, Menschen und Maschinen beizubringen ☆



35 / 5000



[Send feedback](#)

NLP Applications

➤ Question Answering/Comprehension

Passage Sentence

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under gravity.

Question

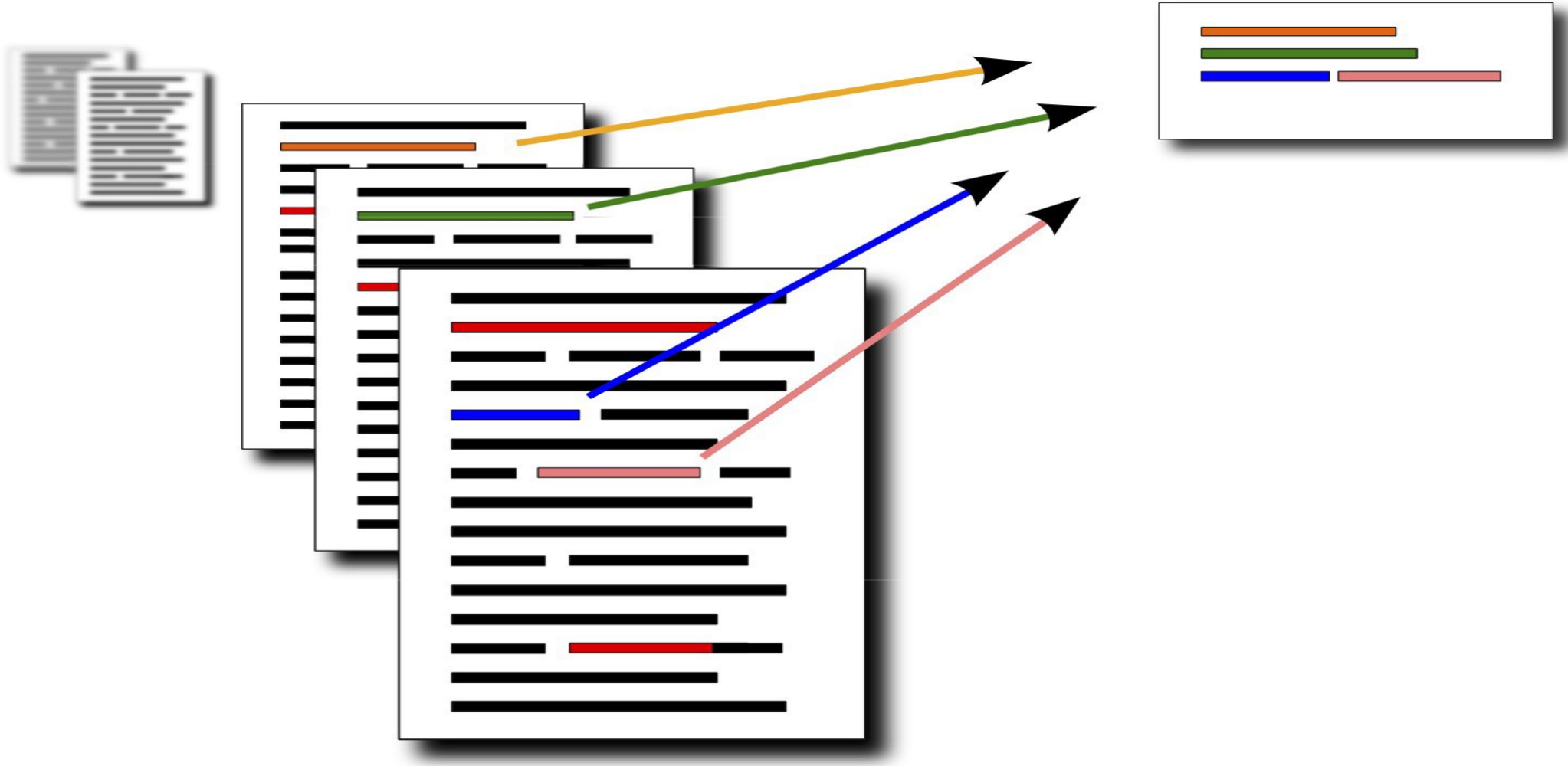
What causes precipitation to fall?

Answer Candidate

gravity

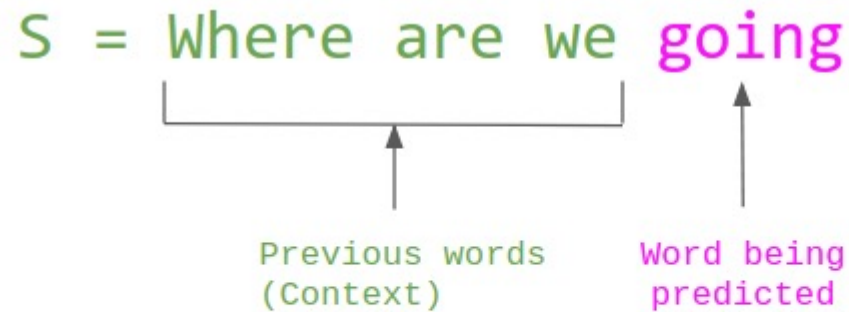
NLP Applications

➤ Text Summarization



Language Modelling

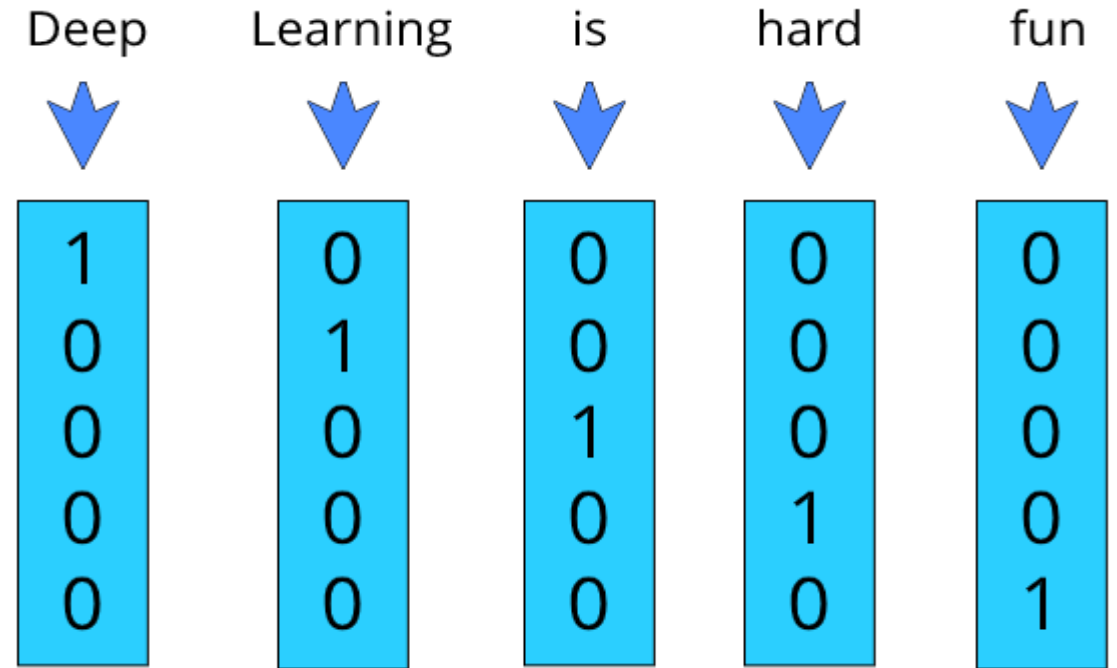
- A probability distribution over words or word sequences
- Probability distribution over strings of text
 - How likely is a string in a given “language”?



$$P(S) = P(\text{Where}) \times P(\text{are} \mid \text{Where}) \times P(\text{we} \mid \text{Where are}) \times P(\text{going} \mid \text{Where are we})$$

Representing text: One-hot Encoding

- One-hot vector of an ID is a vector filled with 0s, except for a 1 at the position associated with the ID.
 - For e.g., for vocabulary size $D=10$, the one-hot vector of word (w) having $ID=4$ is $e(w) = [0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 0]$
- One-hot encoding makes no assumption about word similarity
 - All words are equally similar/different from each other
- This is a natural representation to start with, though a poor one



Representing text: Count Vectorizer

- Counts the number of occurrences of each word appearing in a document
- Converts a collection of text documents to a vector of term/token counts

	the	red	dog	cat	eats	food
1. the red dog →	1	1	1	0	0	0
2. cat eats dog →	0	0	1	1	1	0
3. dog eats food →	0	0	1	0	1	1
4. red cat eats →	0	1	0	1	1	0

Representing text: tf-idf vectorizer

- Stands for Term Frequency – Inverse Document Frequency (tf-idf)
- Common algorithm to transform text into a meaningful representation of numbers

TF*IDF

TF*IDF = TERM FREQUENCY * INVERSE DOCUMENT FREQUENCY

TERM FREQUENCY =
THE AMOUNT OF TIMES A
TERM APPEARS IN A
DOCUMENT

X

INVERSE DOCUMENT
FREQUENCY =
A MEASURE OF WHETHER A
TERM IS RARE OR COMMON
IN A COLLECTION OF
DOCUMENTS.

occurrences of term
in document

↓

$w_{i,j} = t f_{i,j} \times \log \frac{N}{df_j}$

↑

tf-idf score

total
documents

↙

N

↑

documents
containing word

df_j

- This cheesecake is really **yummy**. I'm going for another slice.
- The restaurant service was very slow.

Representing text as vectors

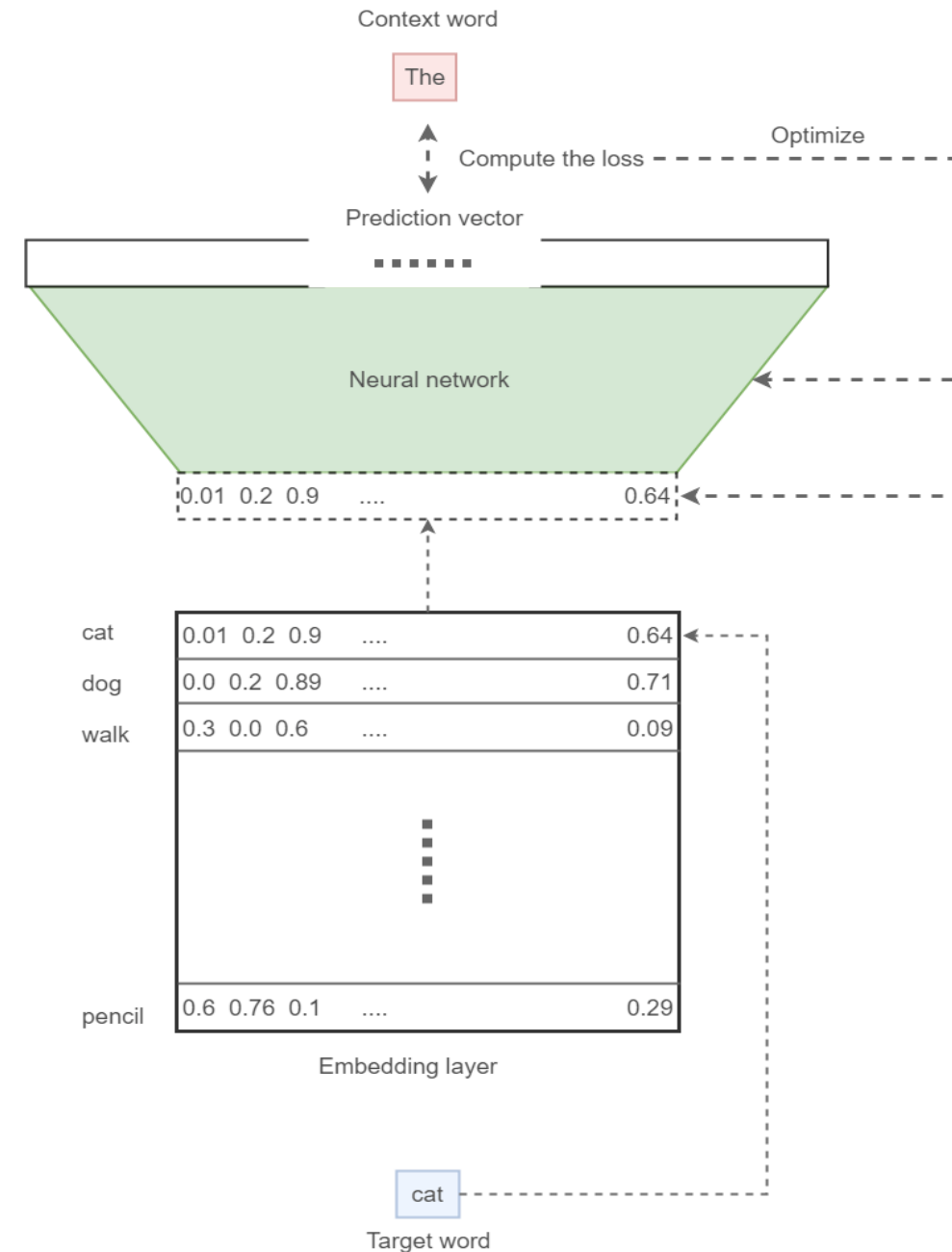
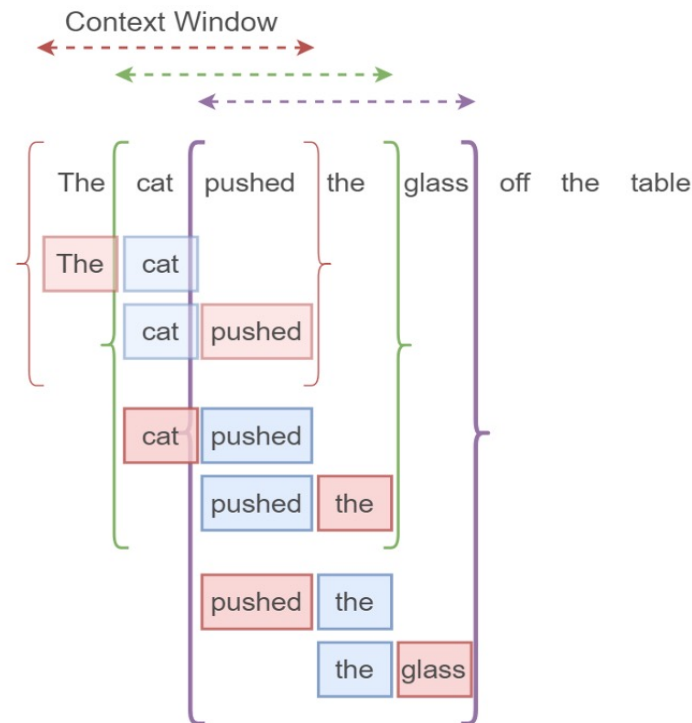
- One-hot encoding, count and tf-idf vectors are **sparse** and **long**
 - Most values are zeros
 - Length of vector is equal to size of vocabulary (can be more than 50,000)
- Alternative is to look for **dense vectors**. These are
 - short and dense
 - easier to use a features in machine learning (less weights to tune)
 - can capture context

Word Embeddings

- Instead of counting how often each word w occurs near “university”, train a classifier on a binary prediction task:
 - Is w likely to show up near “university”?
- We don’t actually care about this task, but we take the learned classifier weights as the word embeddings
- Uses free-form text as implicitly supervised training data
 - A word s near “university” acts as gold standard “ground-truth” to the question
 - No need for hand-labeled supervision
- Word vector algorithms use the context of the words to learn numerical representations for words, so that words used in the same context have similar looking word vectors.

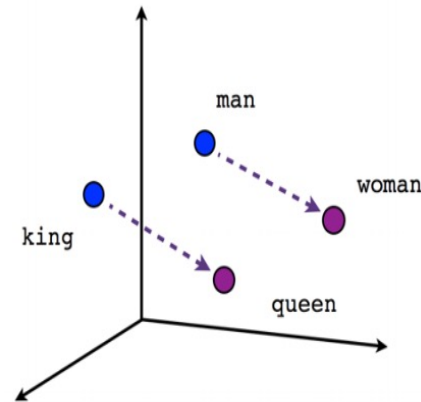
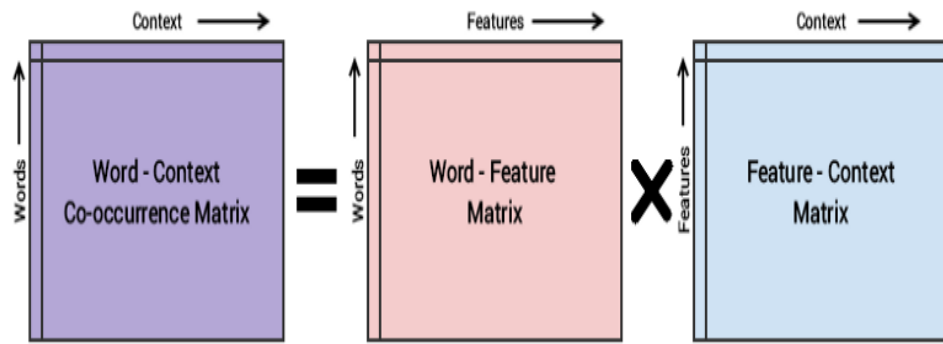
Word2Vec

- “You should know a word by the company it keeps”
– J.R. Firth
- Uses CBOW or Skip-gram model to train the network
- Word2Vec uses skip-gram neural network to predict neighbor context

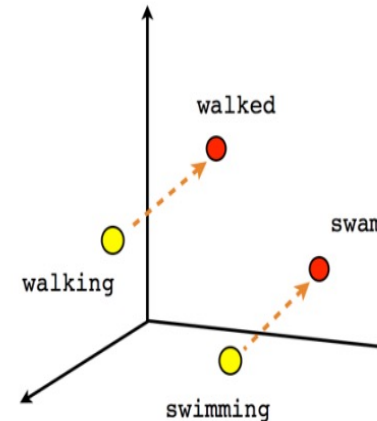


GloVe

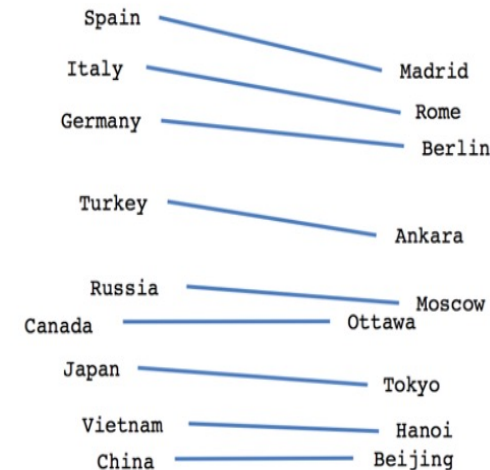
- Word2Vec rely on local statistics (local context information of words)
- GloVe captures both local and global statistics (word co-occurrence) to obtain word vectors
- Can derive semantic relationships between words from the co-occurrence matrix
- Requires upfront pass-through entire dataset



Male-Female



Verb tense



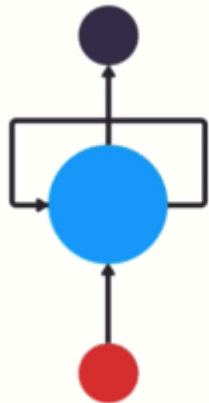
Country-Capital

Statistical Language Modeling

- Models that assign probabilities to the sequences of words
- Focus on n-gram
 - 2-gram (bigram). E.g., “going to”, “on the”
 - 3-gram (trigram). E.g., “what are you”, “please turn your”
- Naïve approach:
 - Calculate the probability of word w , given history, h : $P(w/h)$
 - A computation challenge with significant size of corpus
 - Do approximation using chain rule
- Bigram language model
 - Approximates the probability of a word given all the previous words by using the conditional probability of one preceding word
 - Based on Markov assumption: we can predict the probability of some future unit without looking too far in the past
 - Use Maximum Likelihood Estimation (MLE) to estimate the probability function

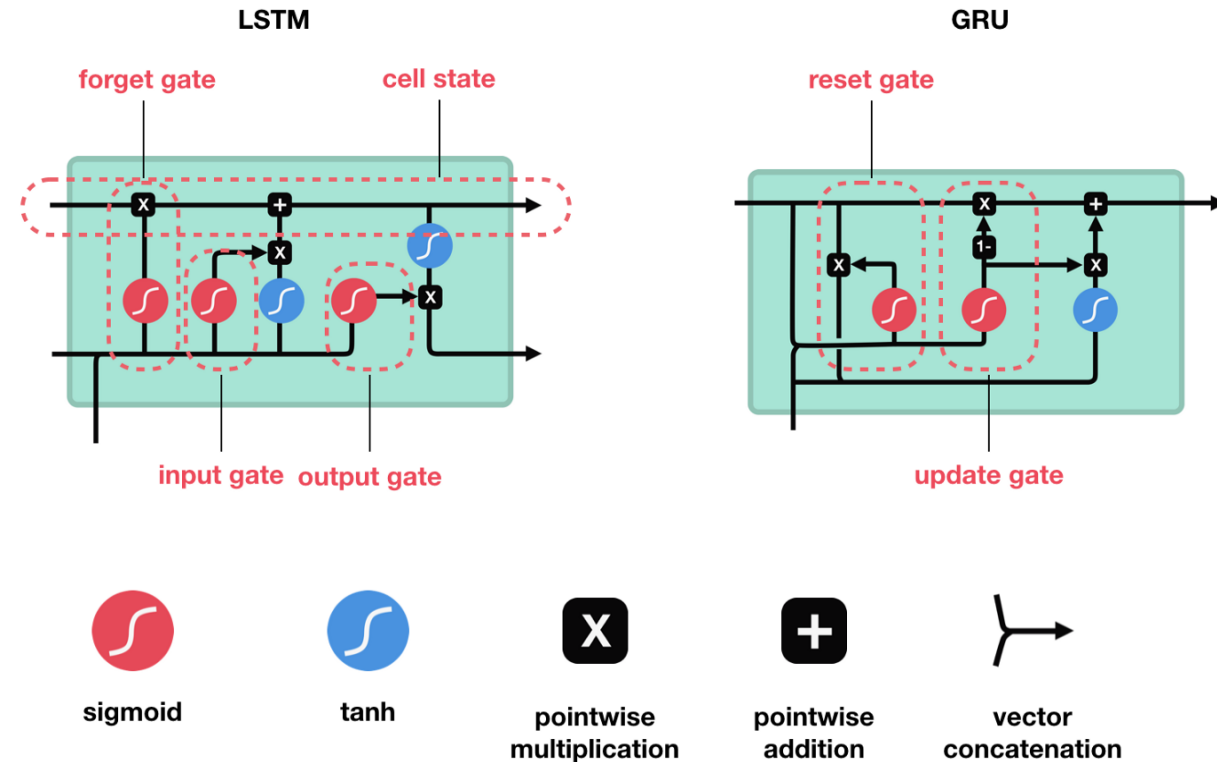
Recurrent Neural Networks (RNNs)

- A class of artificial neural networks suitable for modeling sequential or time-series data
- Exhibit dynamic behavior where connections between nodes form a directed graph along a temporal sequence
- RNNs have a looping mechanism that acts as a highway to allow information to flow from one step to the next. This information is the hidden state, which is a representation of previous inputs.

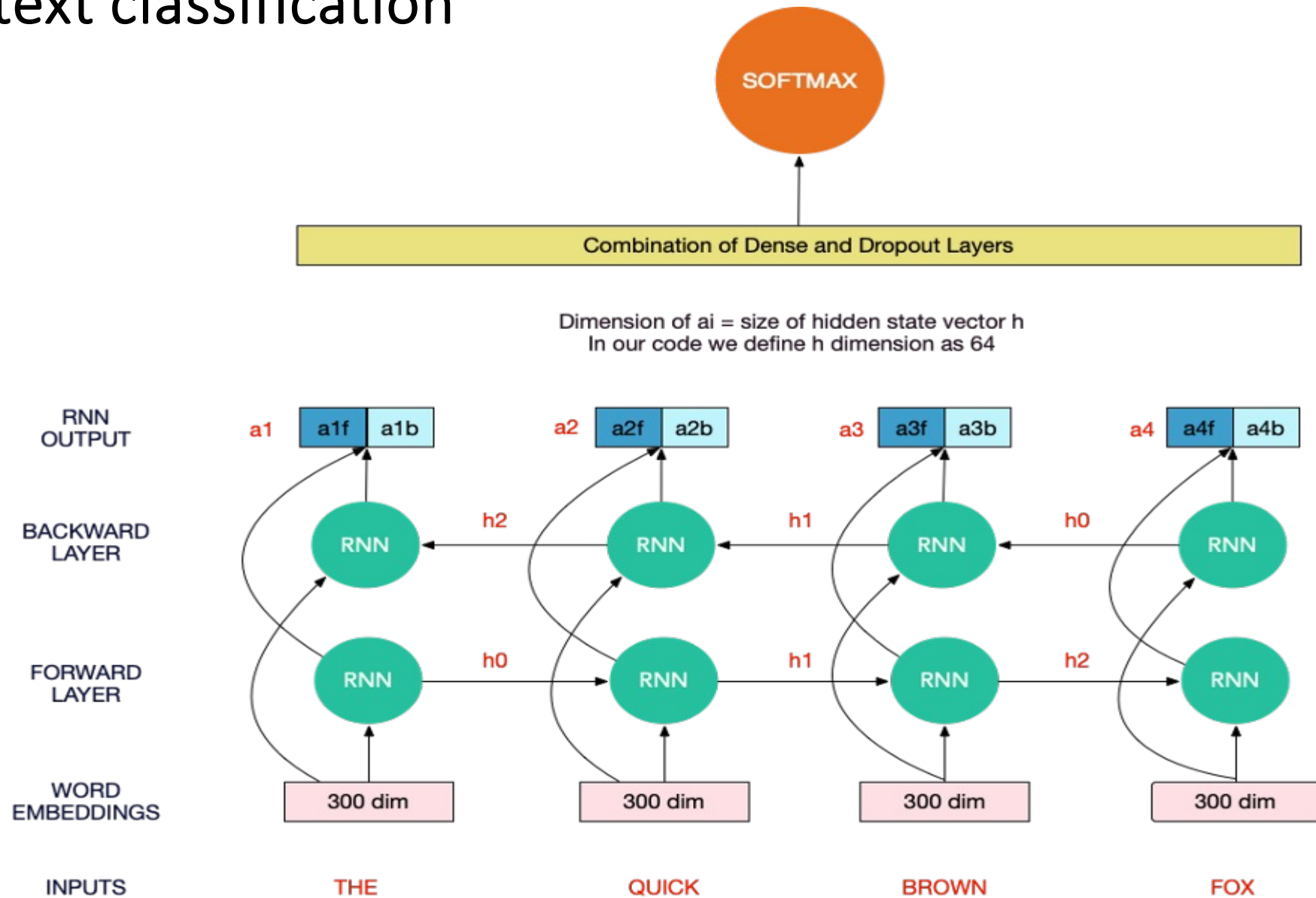


LSTMs/GRUs

- Vanilla RNNs suffers from vanishing gradient problem
 - As the RNNs processes more steps, it has troubles retaining information from previous steps.
 - Due to back-propagation, the earlier layers fail to do any learning as the internal weights are barely being adjusted due to extremely small gradients.
 - Does not learn the long-range dependencies across time steps
- LSTMs and GRUs are two special RNNs, capable of learning long-term dependencies using mechanisms called **gates**.
- These gates are different tensor operations that can learn what information to add or remove to the hidden state.

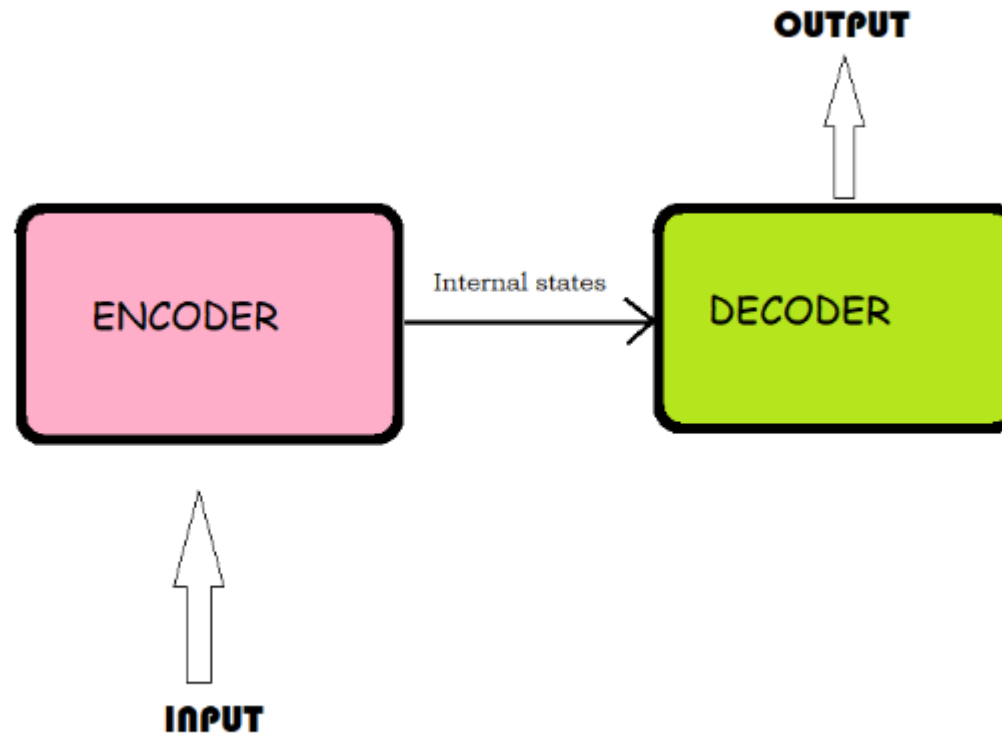


End-to-End text classification



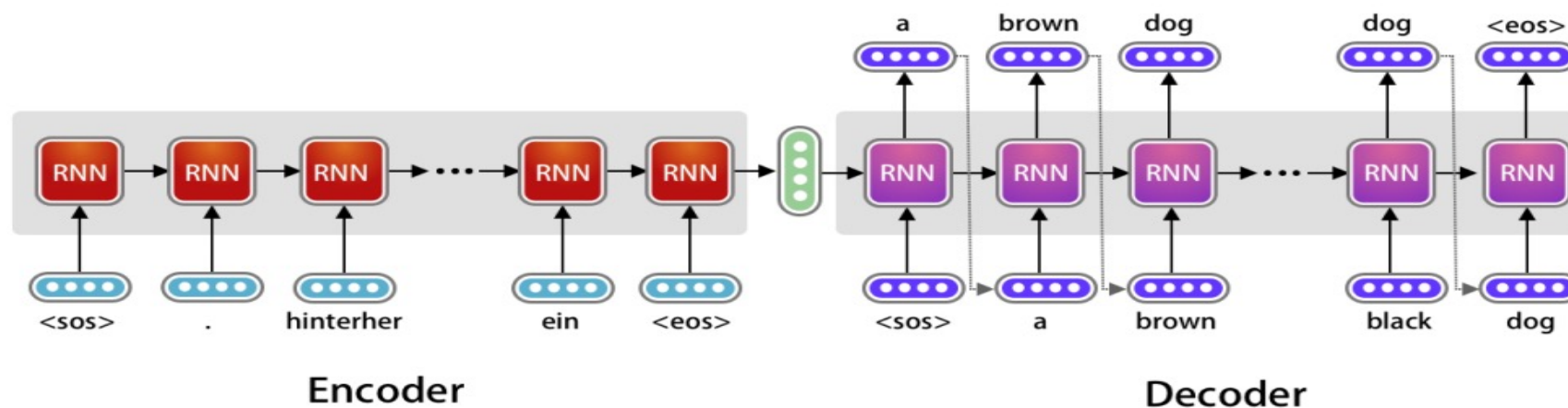
Seq2Seq model

- Model has two parts: Encoder and Decoder (Encoder-Decoder Framework)

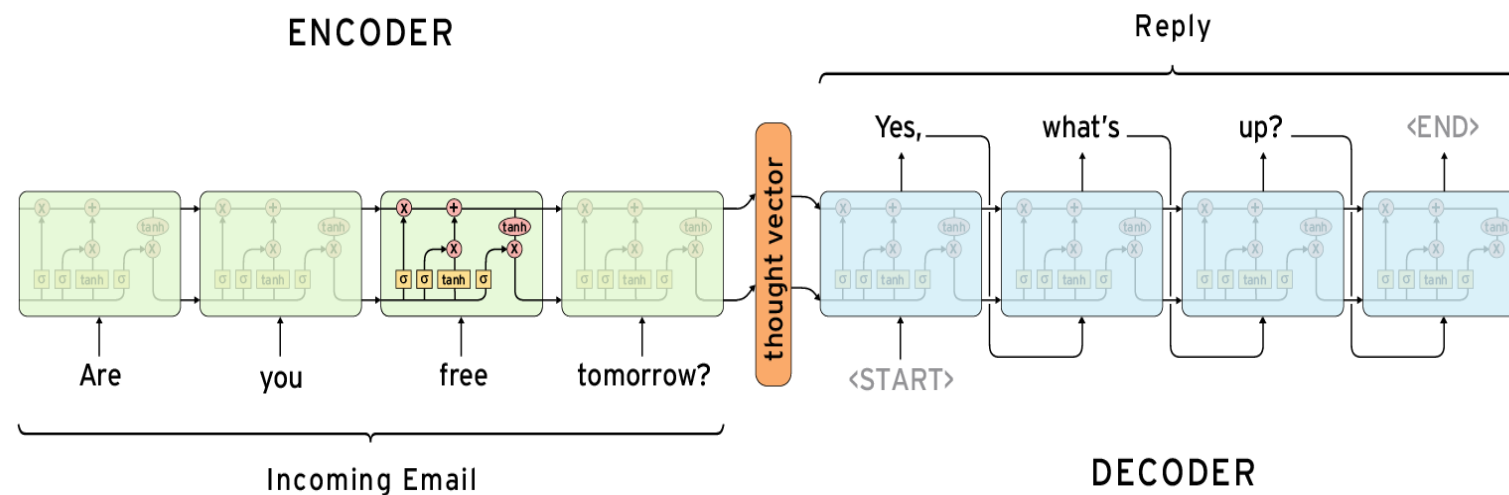


Seq2Seq used for various NLP applications

➤ Machine Translation

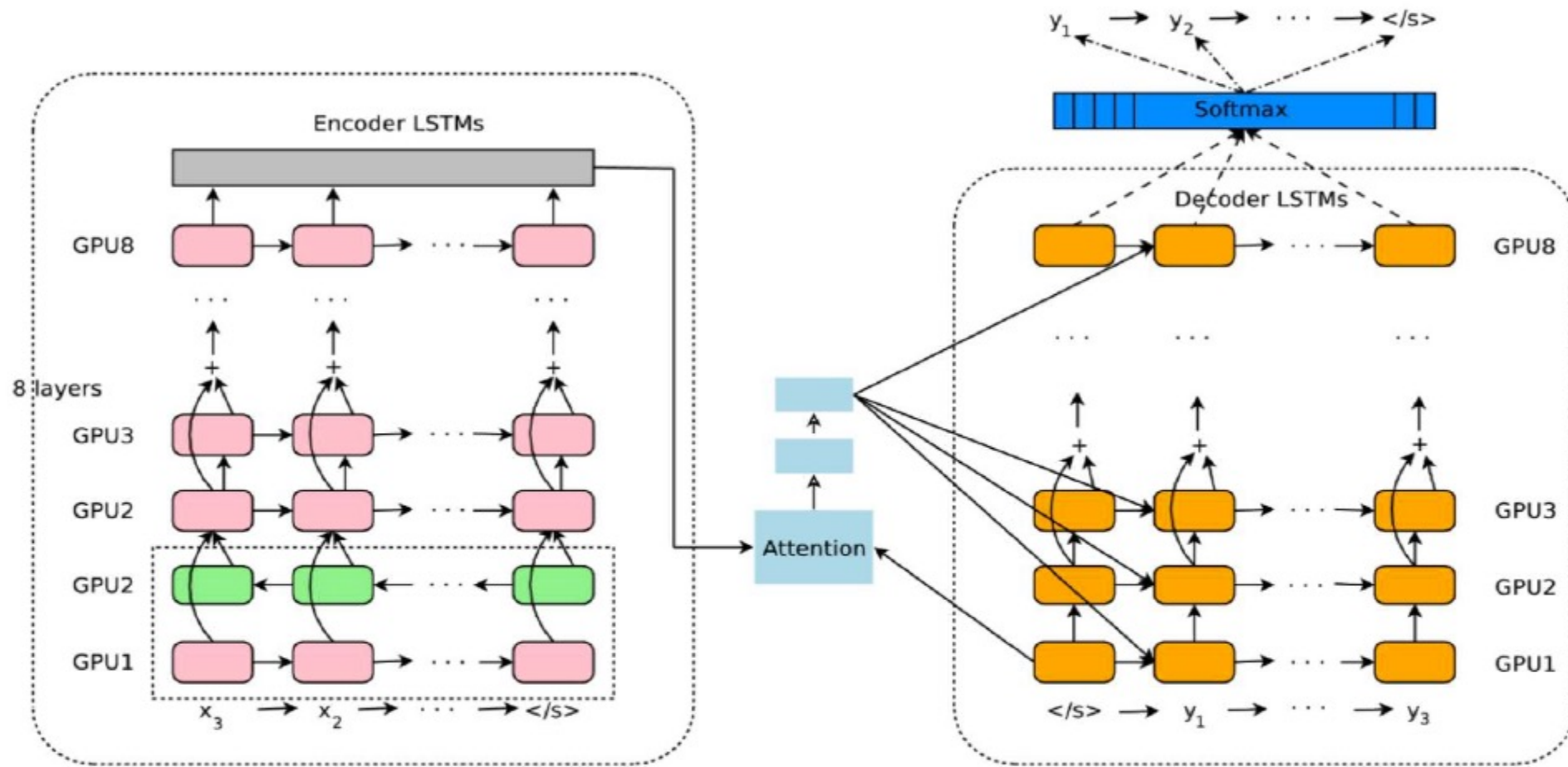


➤ Automatic Email Reply



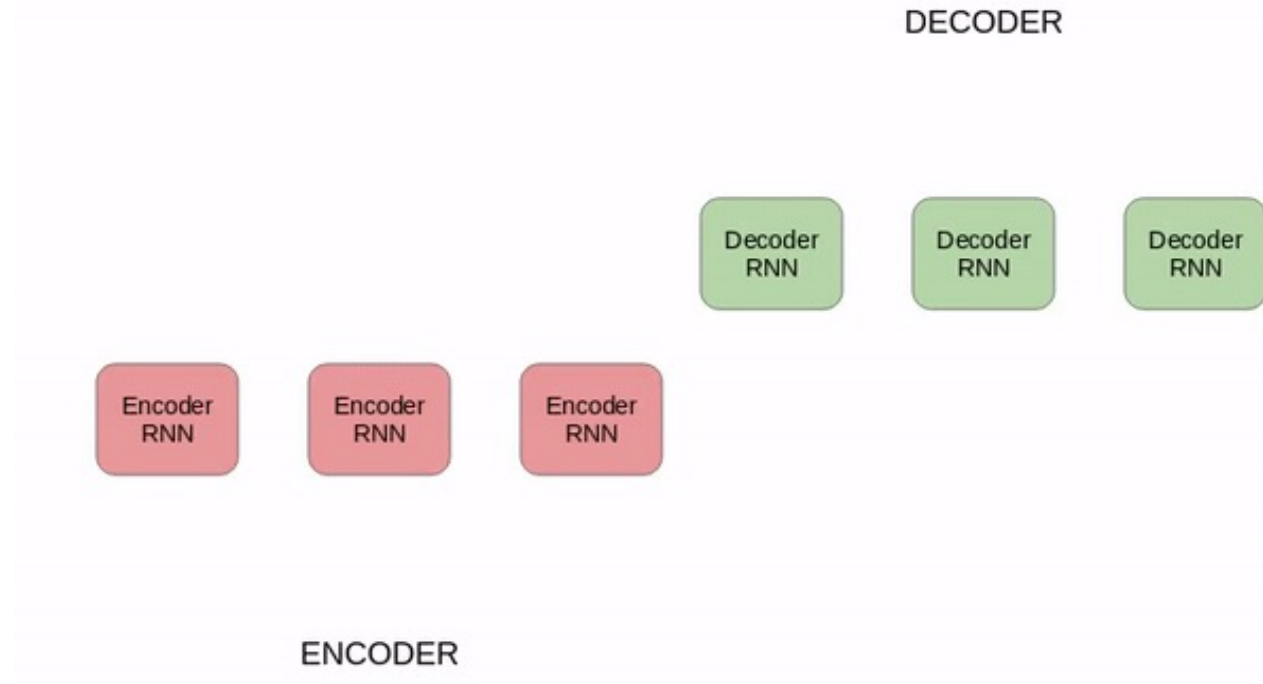
Seq2Seq used for various NLP applications

- Google's Multilingual Neural Machine Translation



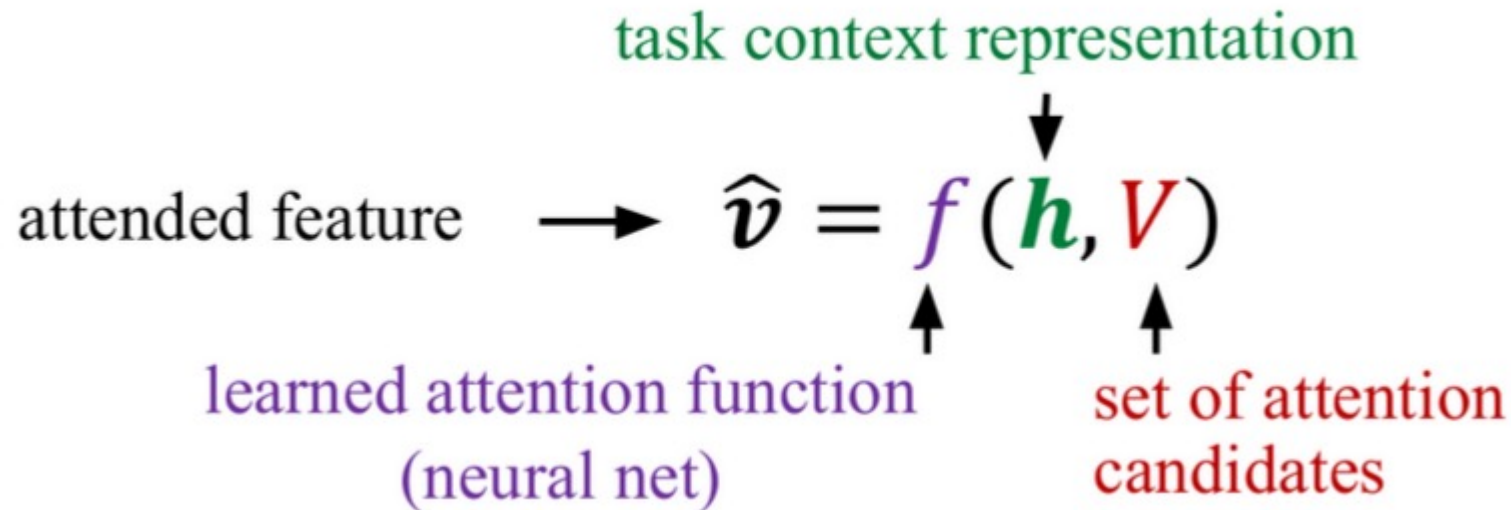
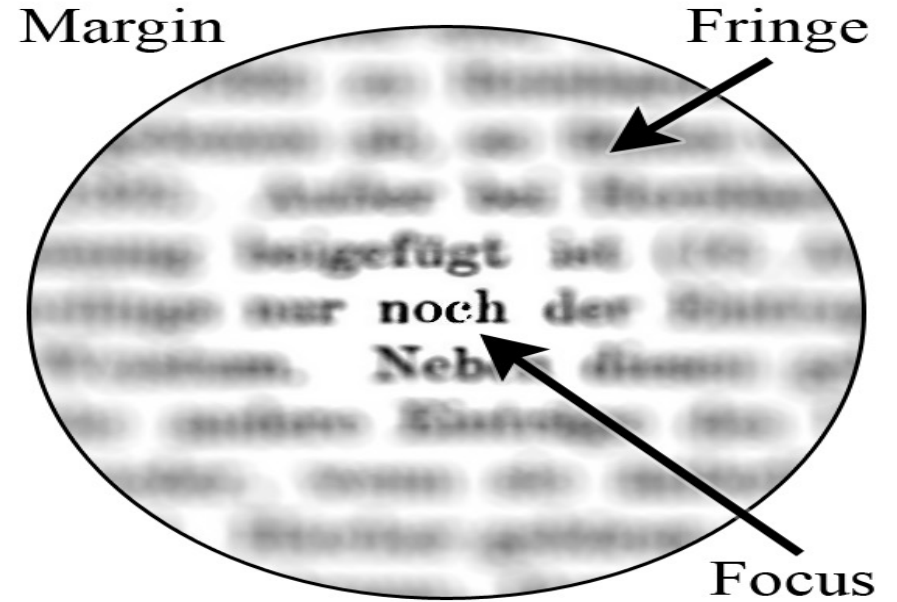
Issues with RNNs for seq2seq tasks

- Despite being very successful for various NLP tasks, there were challenges:
 - dealing with long-range dependencies
 - the sequential nature of the architecture prevents parallelization
- Attention mechanism helped to overcome first issue to certain extent



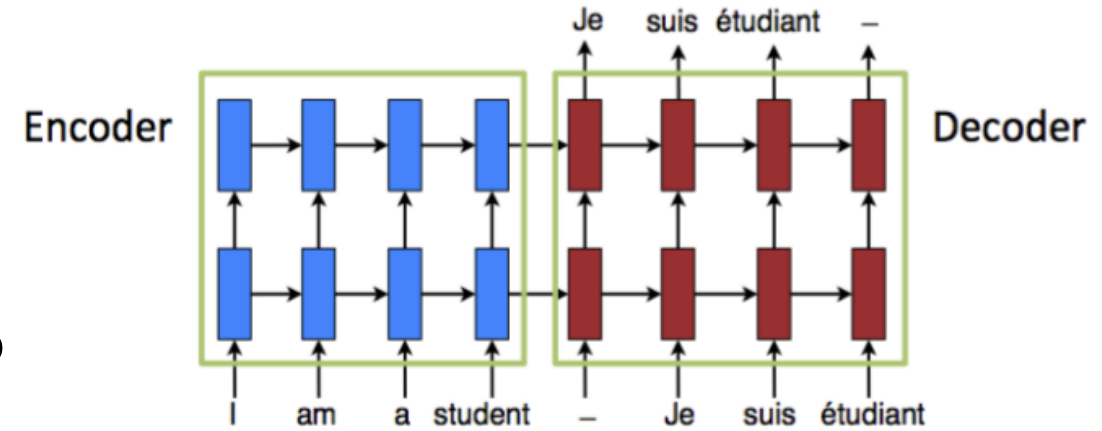
Attention Mechanism

- A set of mechanisms that limit some processing to a subset of incoming stimuli (reducing computational demands)
- Attention in neural networks
 - A mechanism that **learns to focus** on a subset of the **input** that is **relevant to the task**.



Neural Machine Translation by jointly Learning to Align and Translate

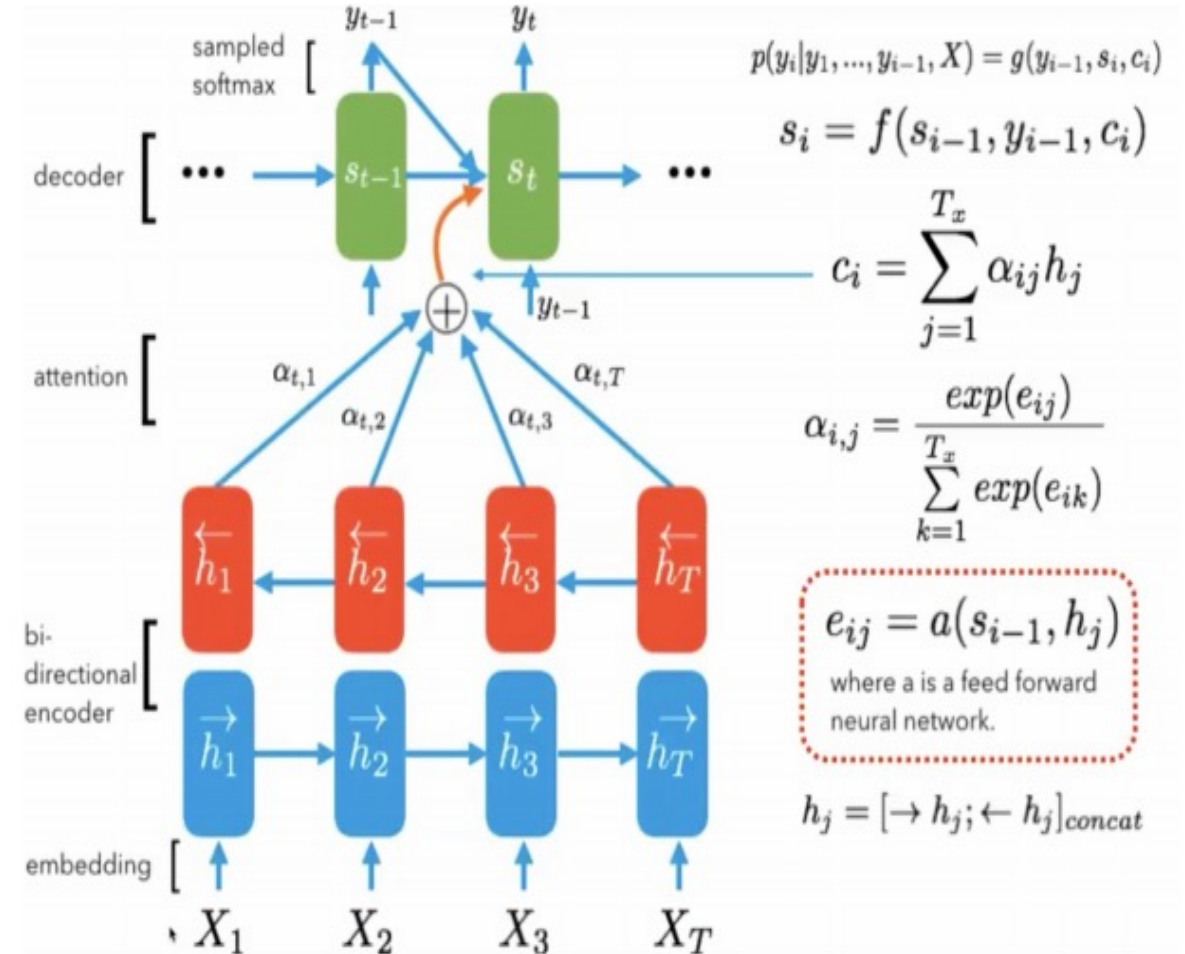
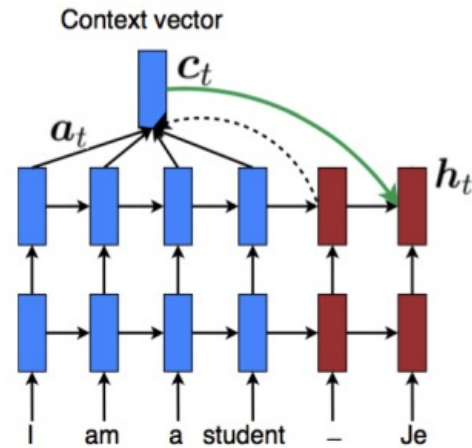
- NMT attempts to build and train a single, large neural network that reads a sentence and outputs a correct translation.
- The NMT system is a seq2seq model (Encoder-Decoder framework)
- Issue: The encoder-decoder framework compresses all the necessary information of a source sentence into a fixed-length vector.
- Cho et al. (2014) showed that the performance of a basic encoder-decoder deteriorates as the length of an input sentence increases.
- Bahdanau et al. (2015) proposed an extension to NMT which learns to align and translate jointly.
- The basic idea of “attention” is that at each time the model generates a word in a translation, it searches for a set of positions in a source sentence where the most relevant information is concentrated
- The model predicts a target word based on the context vectors associated with these source positions and all the previous generated target words



Neural Machine Translation by jointly Learning to Align and Translate

- Solution: Enable the network to pay attention to specific areas of the input by adding new (weighted) connections

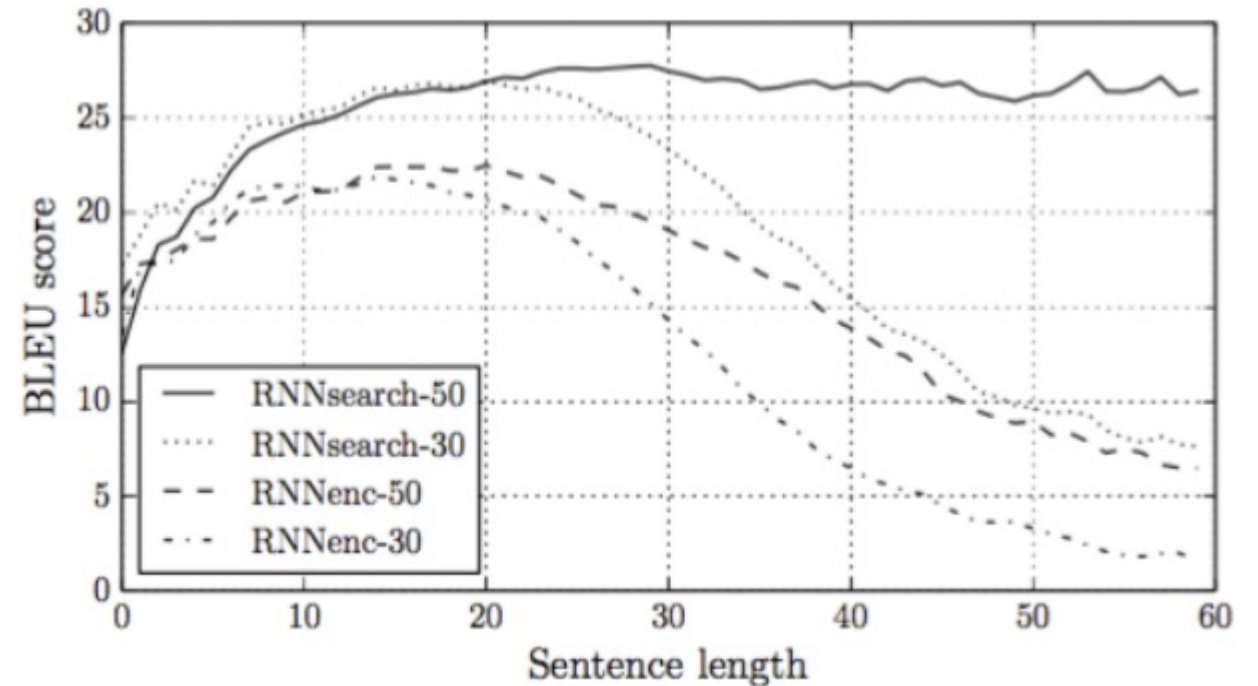
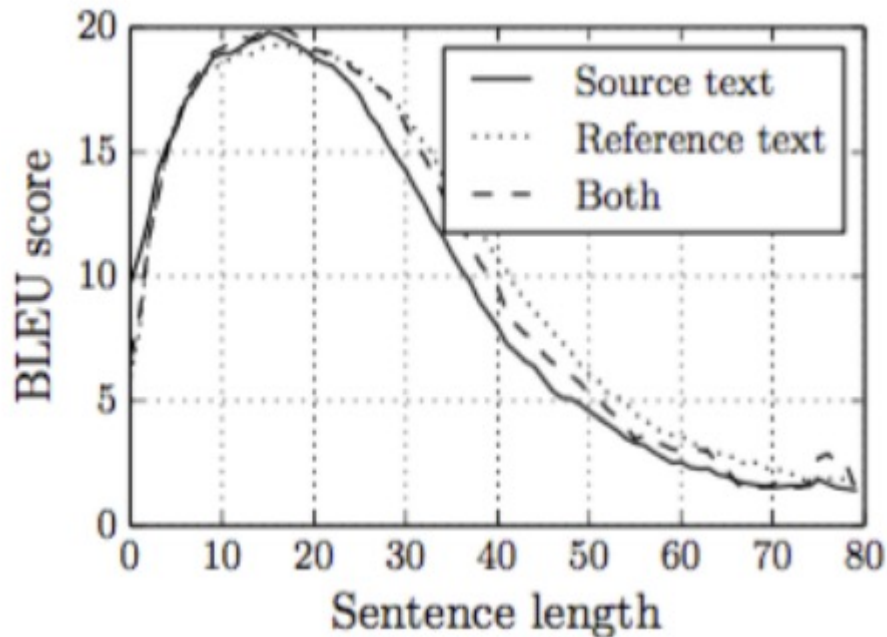
$$c_i = \sum_{j=1}^{T_x} \alpha_{ij} h_j$$
$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^{T_x} \exp(e_{ik})},$$
$$e_{ij} = a(s_{i-1}, h_j)$$



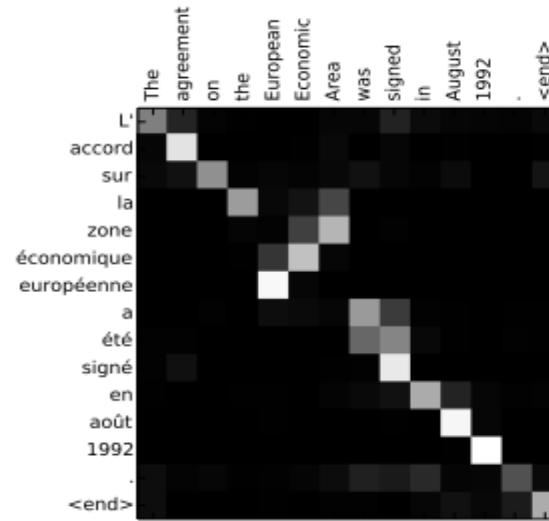
NMT by jointly Learning to Align and Translate

Before Attention: Long sentences are very hard as they are compressed to a fixed length vector

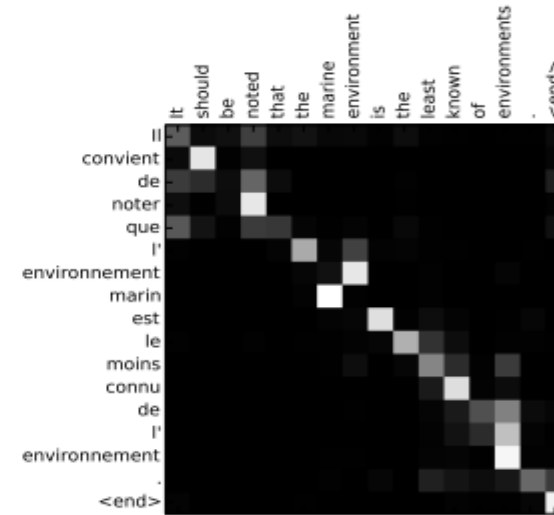
After Attention: The attention mechanism helps to overcome the issue



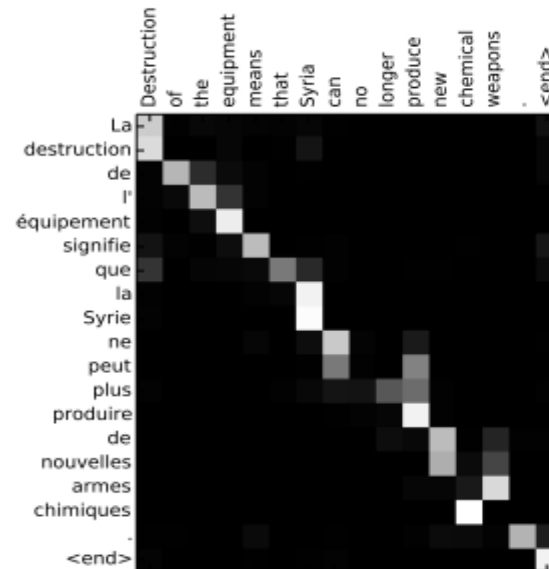
NMT by jointly Learning to Align and Translate



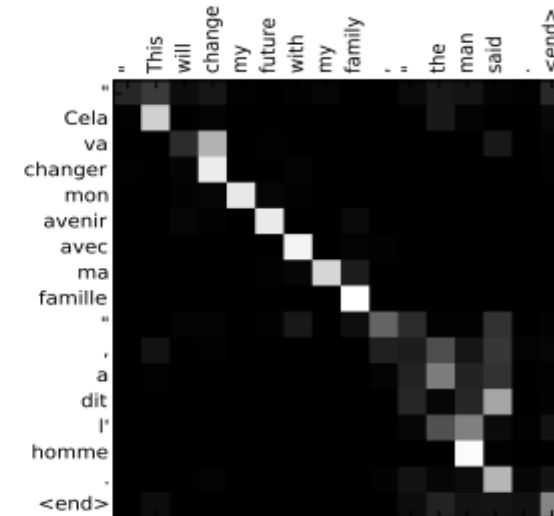
(a)



(b)



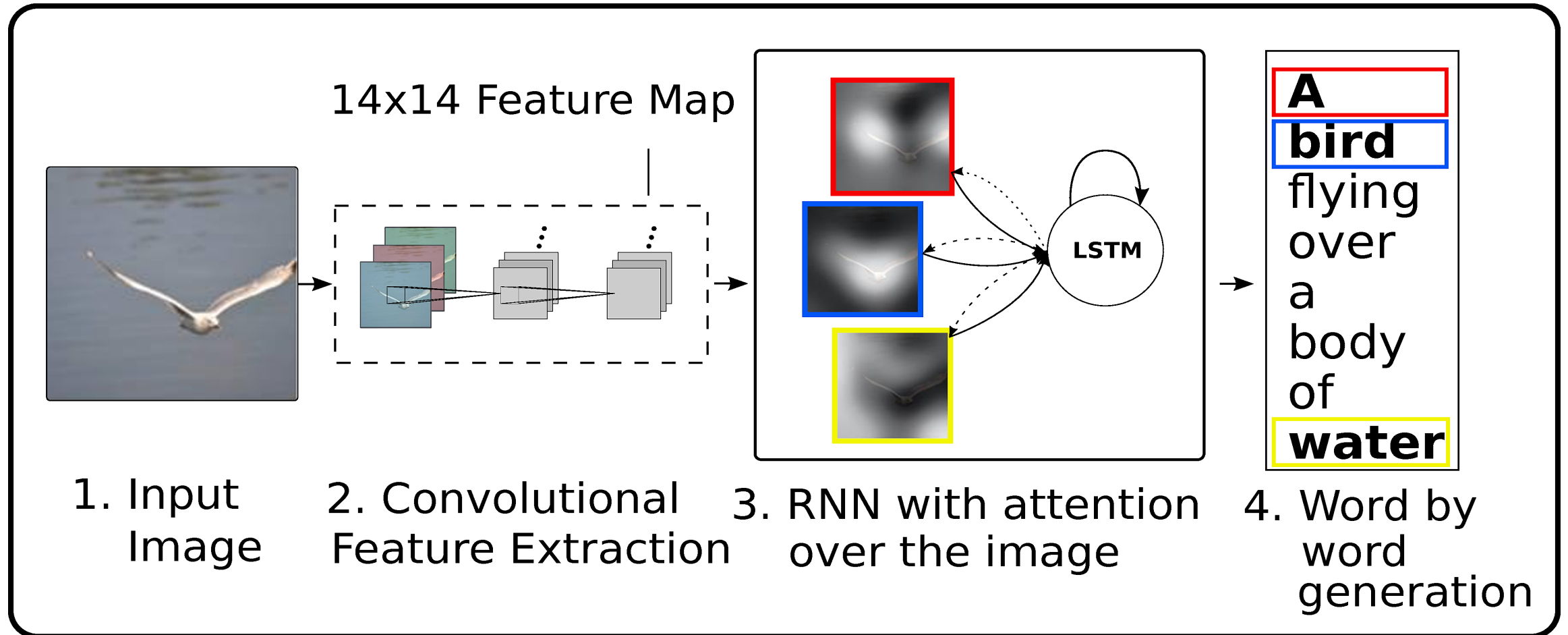
(c)



(d)

Show, Attend and Tell (Xu et al., 2015)

- “Rather than compressing an entire image into static representation, attention allows for **salient features** to dynamically come to forefront as needed”



Show, Attend and Tell (Xu et al., 2015)



A woman is throwing a frisbee in a park.



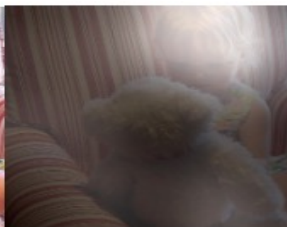
A dog is standing on a hardwood floor.



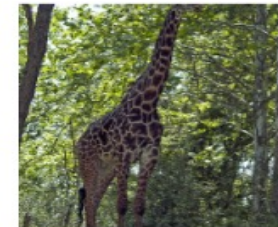
A stop sign is on a road with a mountain in the background.



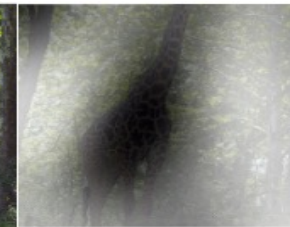
A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.



A giraffe standing in a forest with trees in the background.



<start>



a



large



airplane



flying



in



the



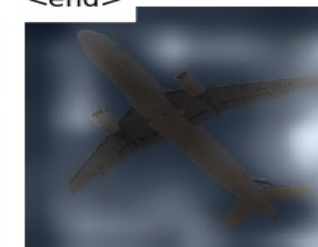
blue



sky



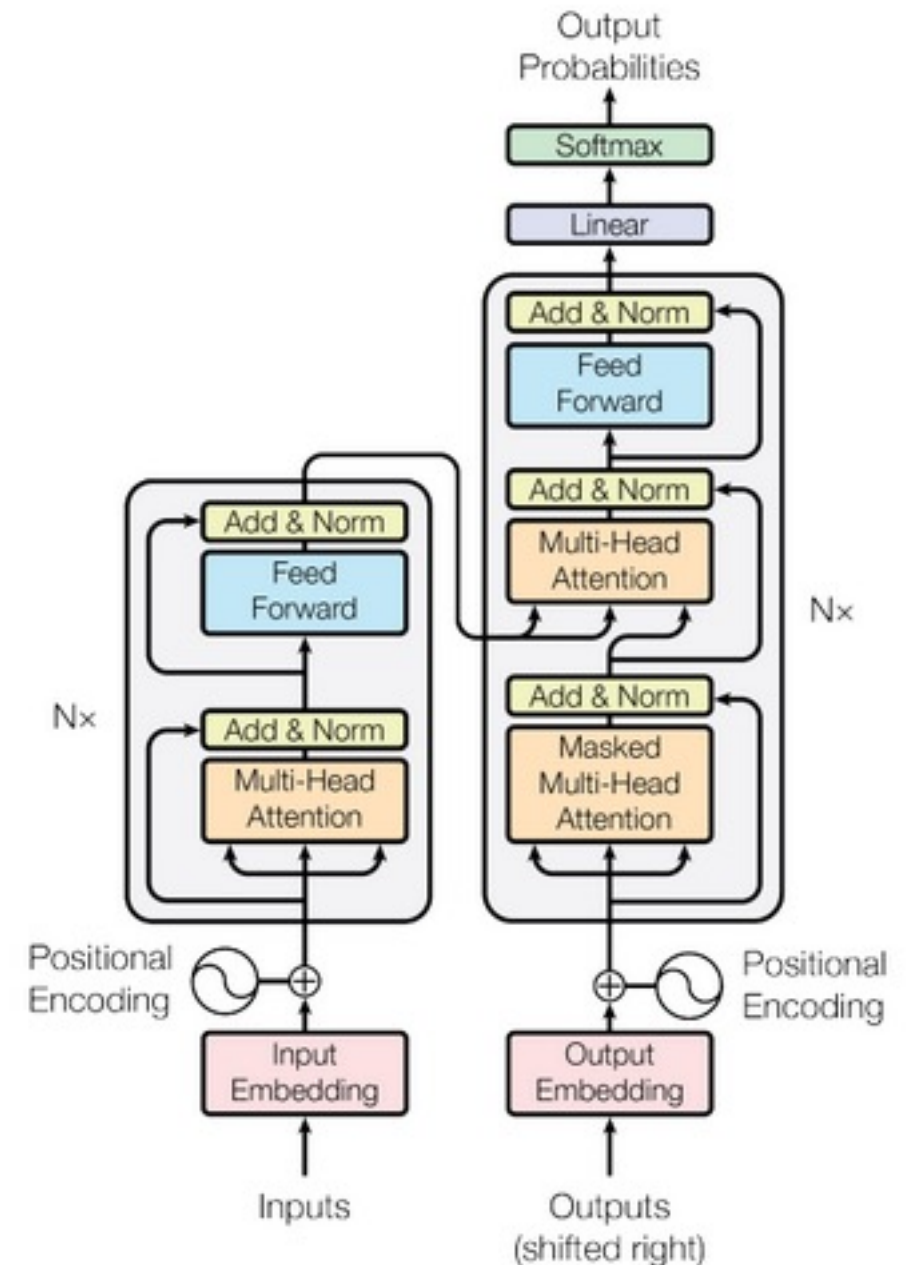
<end>



Transformers

- Attention is All You Need
- Novel architecture relies entirely on **self-attention** to compute representations of its input and output without using sequential RNNs or convolutions.
- Aim is to solve seq2seq tasks while handling long-range dependencies

“Griezmann’s announcement comes as a bit of a shock. After enduring the drama surrounding his potential last summer, many thought he was committed to Atletico for more than a year, but the Frenchman seems to have changed his mind.”

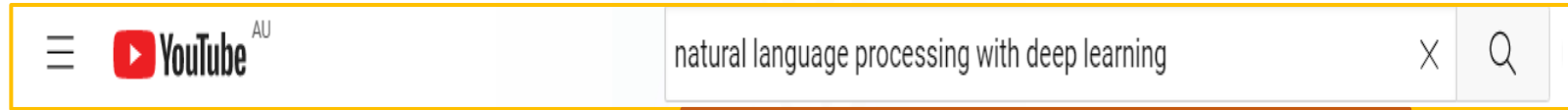


Self-attention in transformers

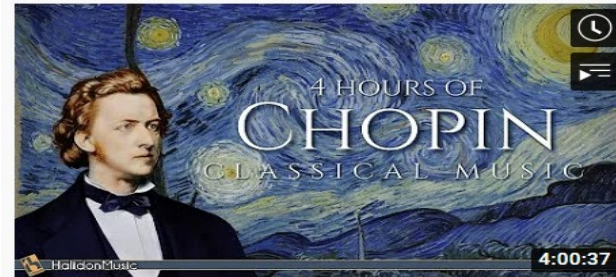
➤ Understanding self-attention with Search

- Query (Q)
- Key (K)
- Value (V)

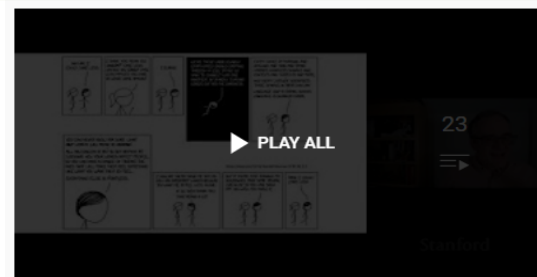
Value (V)



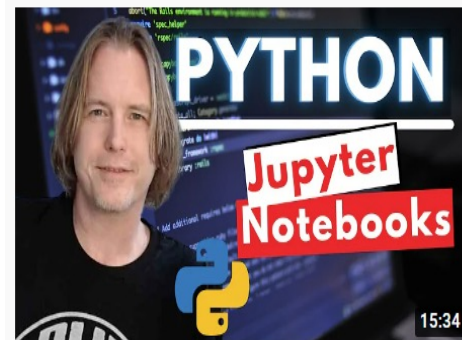
Query (Q)



Key 1 (K1)



Key 2 (K2)



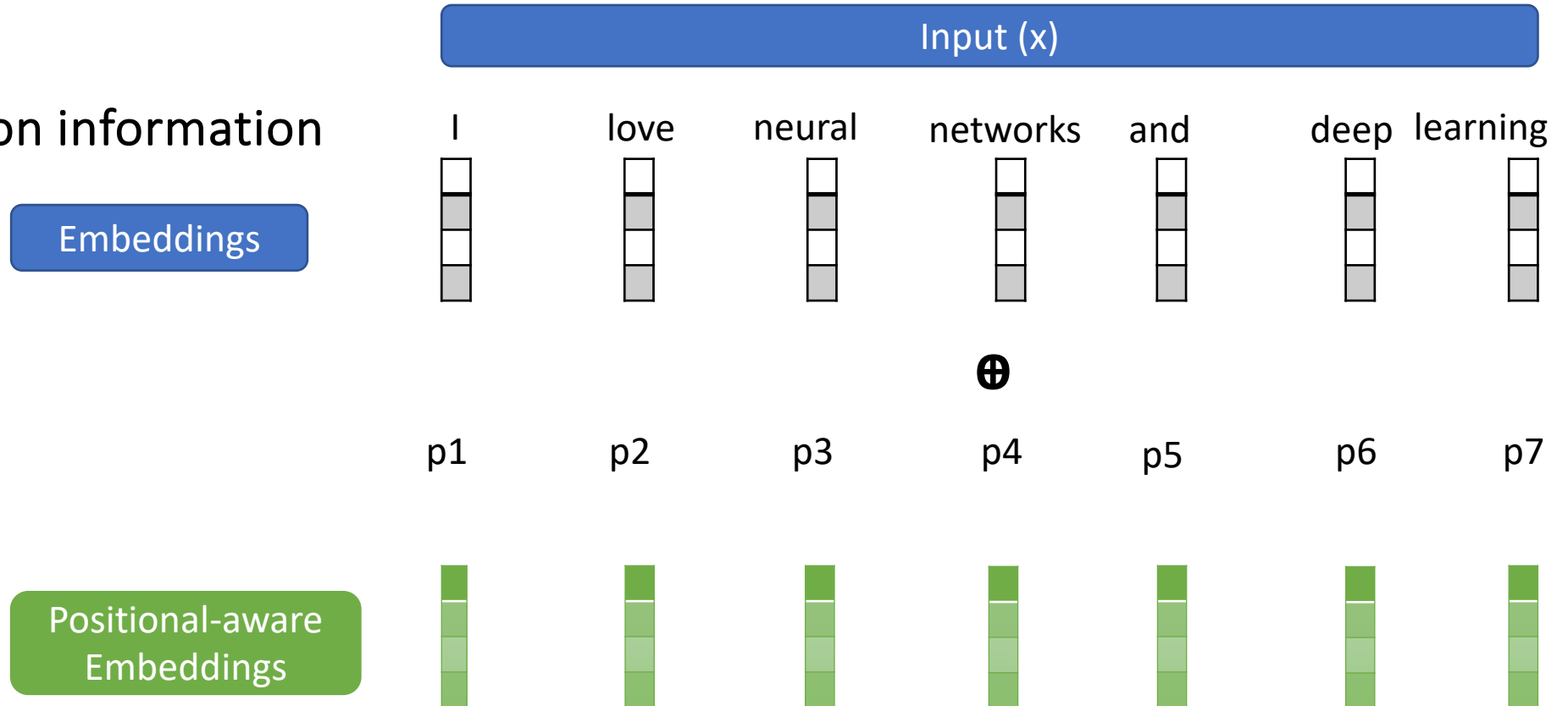
Key 3 (K3)

Self-attention in transformers

Self-attention:

Identify and attend to most important features in input

Step 1: Encode position information



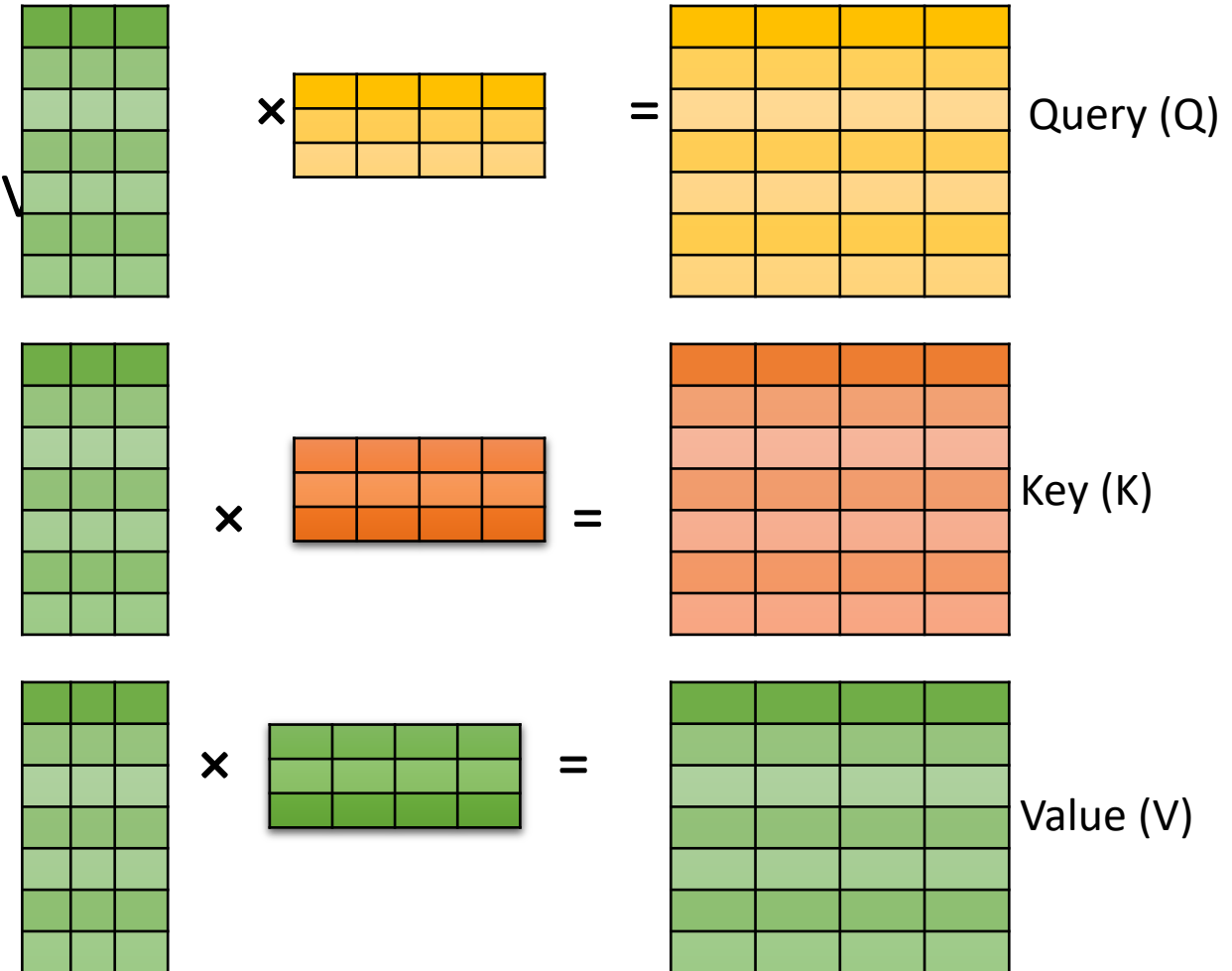
Self-attention in transformers

Self-attention:

Identify and attend to most important features in input

Step 1: Encode position information

Step 2: Extract query (Q), key (K), and value (V)



Self-attention in transformers

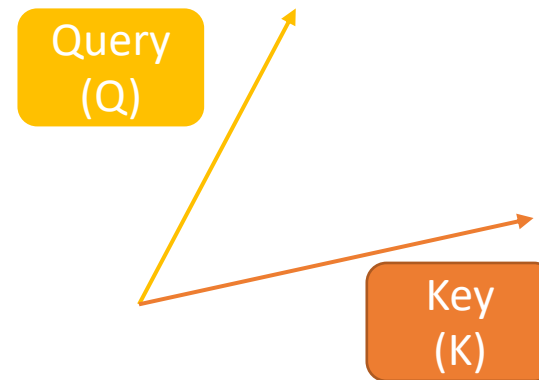
Self-attention:

Identify and attend to most important features in input

Step 1: Encode position information

Step 2: Extract query (Q), key (K), and value (V)

Step 3: Compute attention weighting



$$\frac{Q \cdot K^T}{\text{Scaling}}$$

Self-attention in transformers

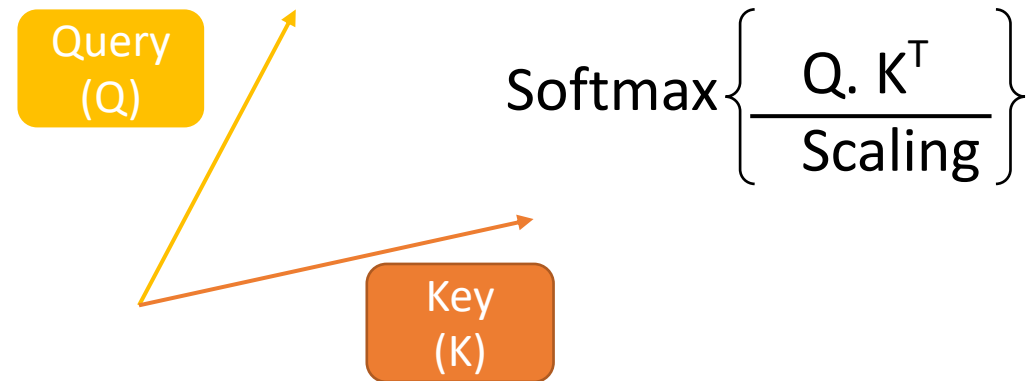
Self-attention:

Identify and attend to most important features in input

Step 1: Encode position information

Step 2: Extract query (Q), key (K), and value (V)

Step 3: Compute attention weighting



Self-attention in transformers

Self-attention:

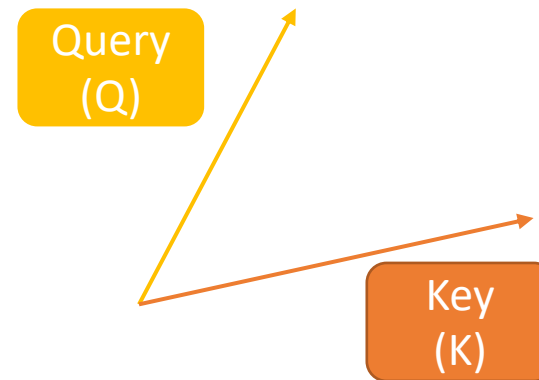
Identify and attend to most important features in input

Step 1: Encode position information

Step 2: Extract query (Q), key (K), and value (V)

Step 3: Compute attention weighting

Step 4: Extract features with high attention

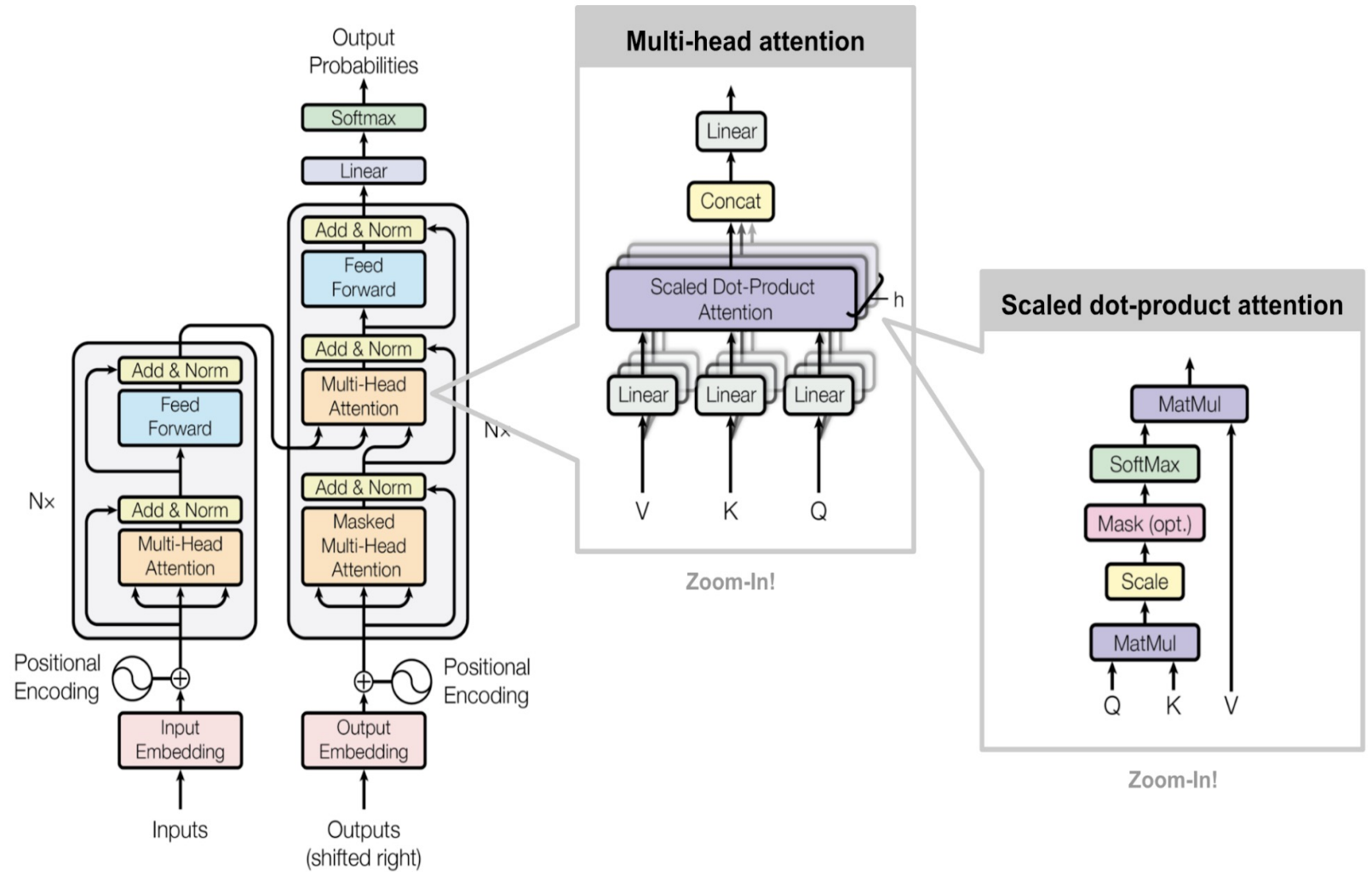


$$\text{Softmax} \left\{ \frac{Q \cdot K^T}{\text{Scaling}} \right\} \times V = A(Q, K, V)$$

Self-attention in transformers

Self-attention:

- Step 1: Encode position information
- Step 2: Extract query (Q), key (K), and value (V)
- Step 3: Compute attention weighting
- Step 4: Extract features with high attention



BERT

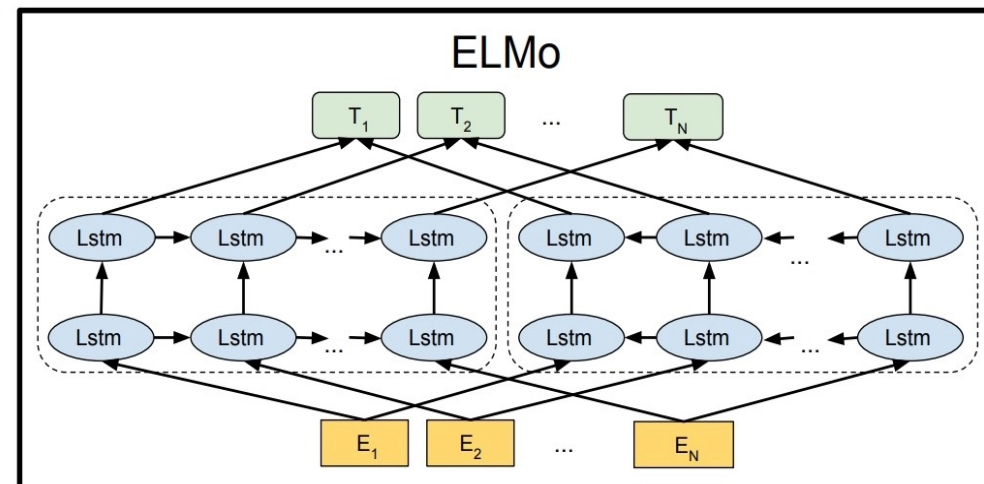
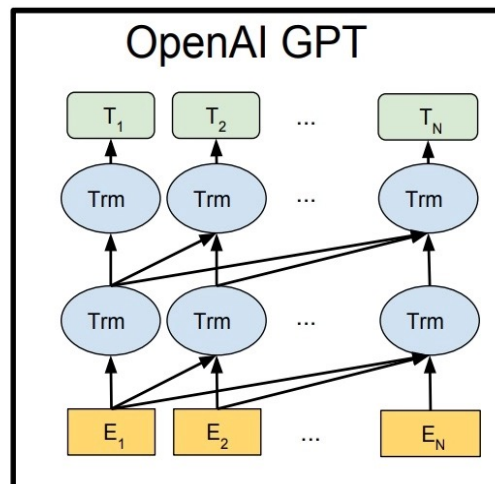
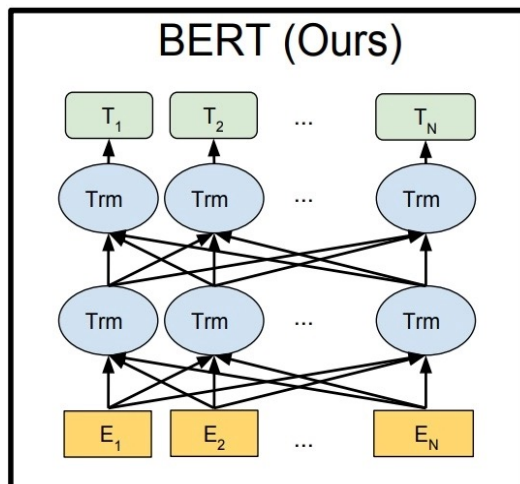
- Bidirectional Encoder Representations from Transformers
- Idea to have **deeply bidirectional, unsupervised language representation**
- Example sentence: “I accessed the bank account”
 - Unidirectional contextual model would represent “bank” based on “I accessed the”
 - BERT represent “bank” based on both its previous and next context – “I accessed the --- account”
- Why bidirectional in BERT possible?
 - Not possible to train bidirectional models by simply conditioning each word on its previous and next words, since this would allow the word that’s being predicted to indirectly “see itself” in a multi-layer model.
 - BERT rely on two tasks
 - Masking words in the input
 - Next sentence prediction

Input: The man went to the [MASK]₁ . He bought a [MASK]₂ of milk .
Labels: [MASK]₁ = store; [MASK]₂ = gallon

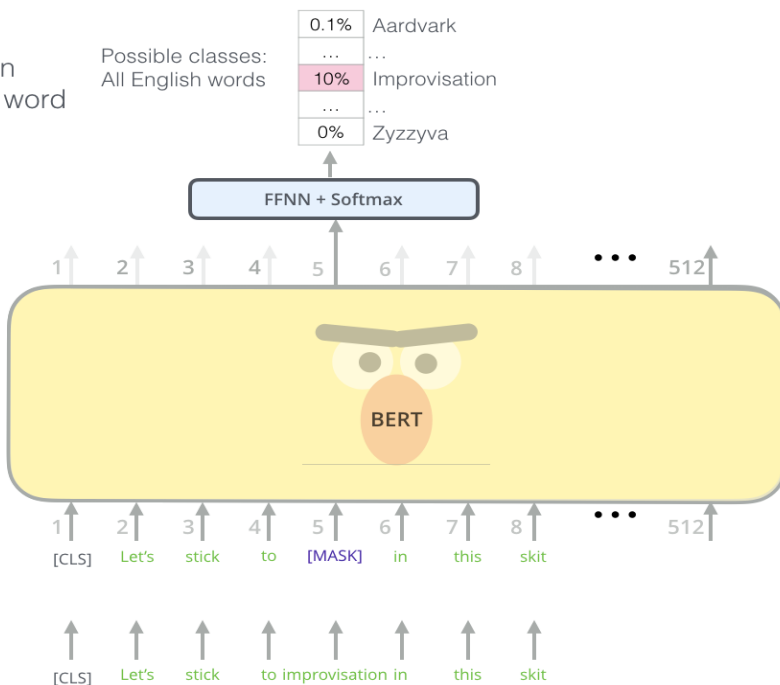
Sentence A = The man went to the store.
Sentence B = He bought a gallon of milk.
Label = IsNextSentence

Sentence A = The man went to the store.
Sentence B = Penguins are flightless.
Label = NotNextSentence

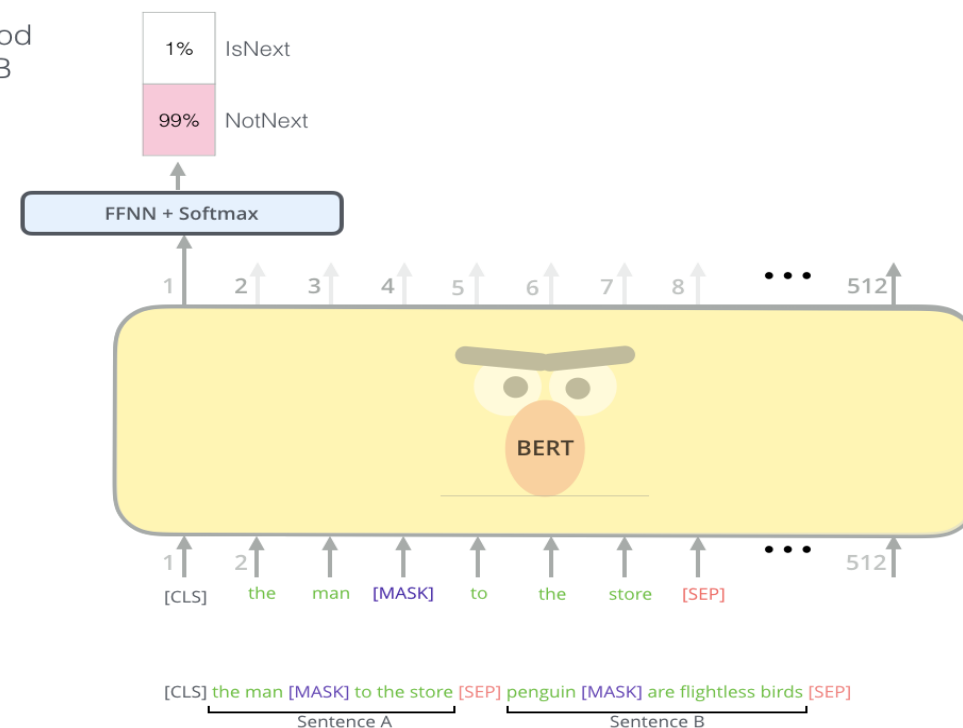
BERT



Use the output of the masked word's position to predict the masked word

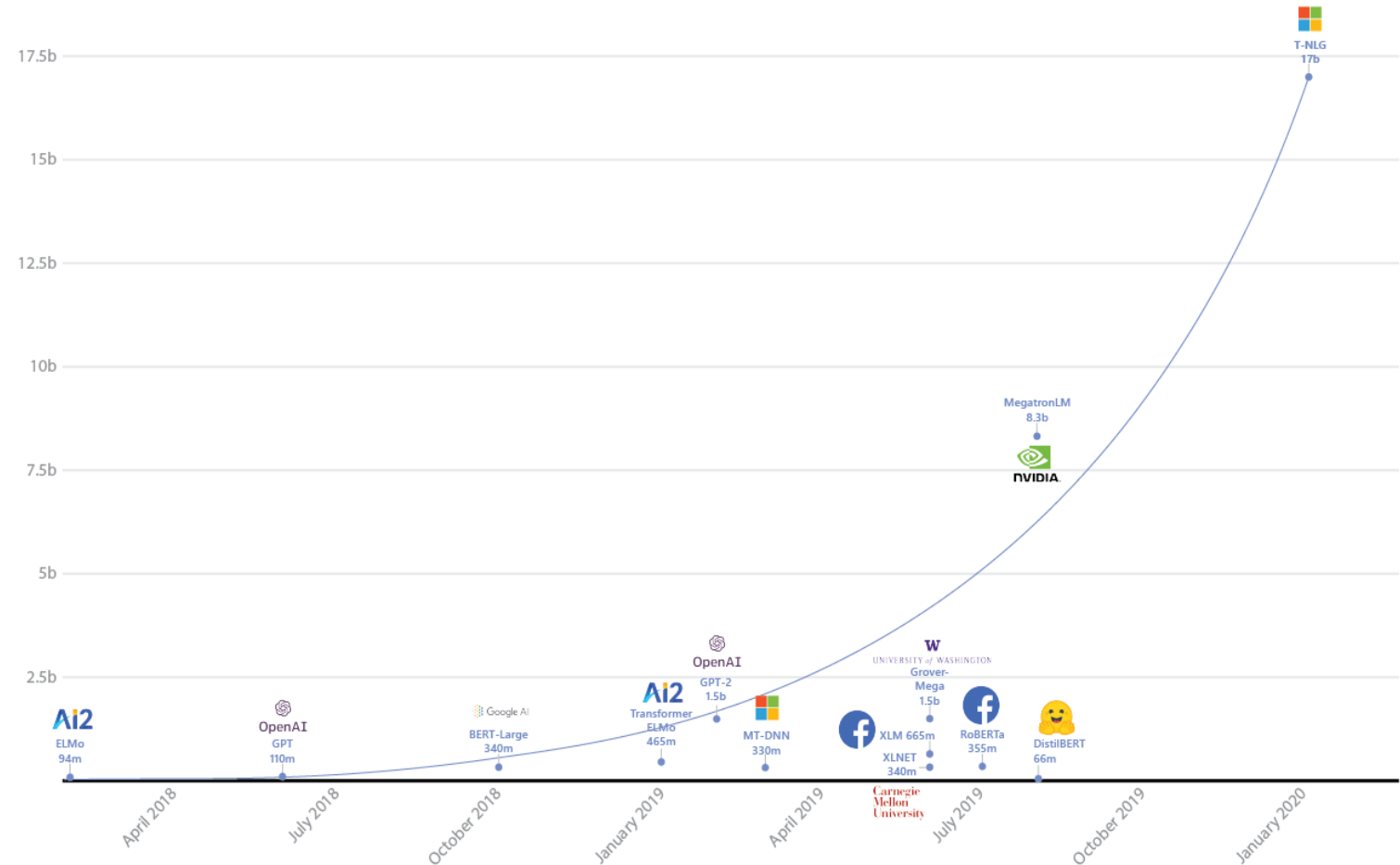


Predict likelihood that sentence B belongs after sentence A



Turing-NLG

- A Transformer based generative language model
- Based on DeepSpeed (DL library) to make distributed training of large models easier
- 17 billion parameters



OpenAI GPT-3

- Key idea: Improve task-agnostic few-shot performance
- Evaluation on various NLP tasks under few-shot learning, one-shot learning, and zero-shot learning demonstrates GPT-3 promising results
- Technical details:
 - An autoregressive language model
 - GPT-3 has 96 layers with each layer having 96 attention heads
 - trained on datasets with 500 billion tokens
 - Word embedding size of 12888
 - Context window size is of 2048 tokens
 - Uses alternating dense and locally banded sparse attention patterns
- Compute:
 - Trained on more than 576 GB of text data including common crawl, Books, and Wikipedia
 - About 175 billion parameters
 - costs OpenAI around \$4.6 million

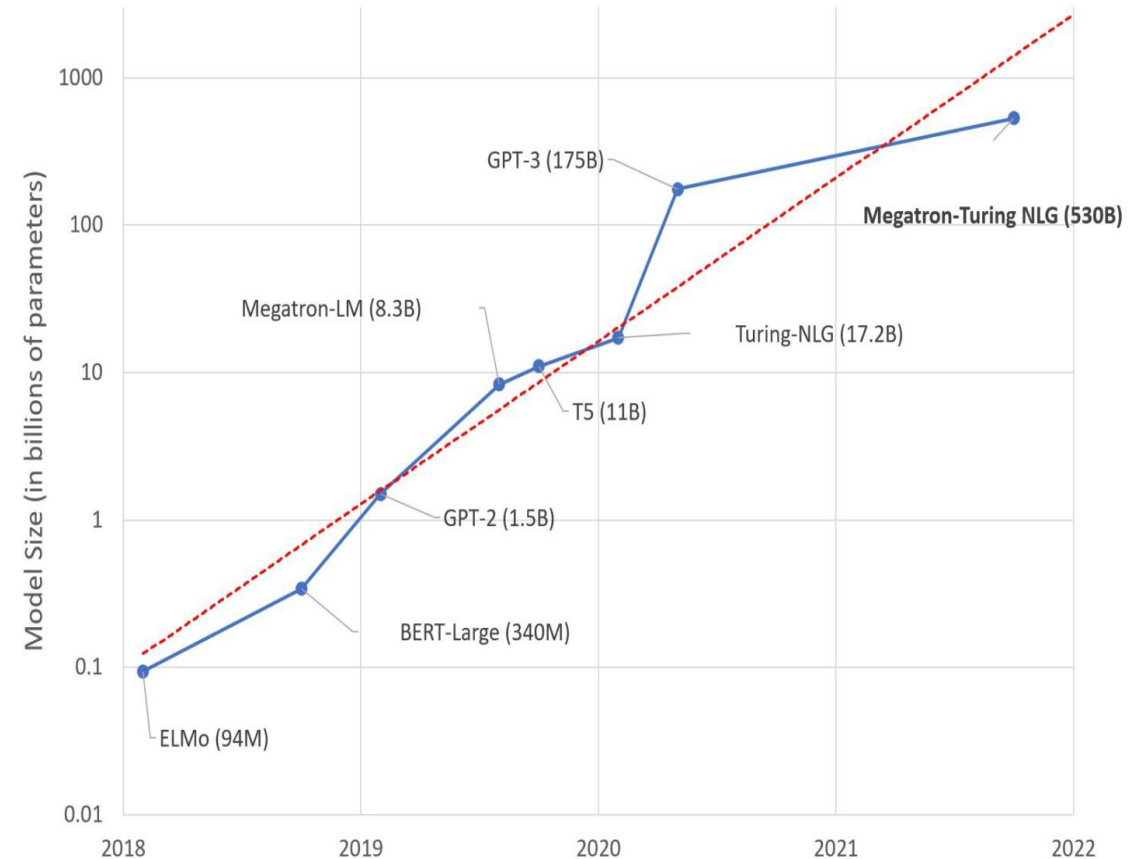
Behold the Megatron: Microsoft and Nvidia build massive language processor

MT-NLG is a beast that fed on over 4,000 GPUs

Katyanna Quach

Tue 12 Oct 2021 // 00:36 UTC

- Largest transformer model to date
- 530 billion parameters
- Trained on a wide variety of NLP tasks such as auto-completing sentences, question and answering, reading and reasoning
- Outperforms various tasks with little to no fine-tuning, in a few-shot or zero-shot learning
- Trained using Nvidia's Selene machine learning supercomputer
- Estimated to cost \$85 million



OpenAI GPT-4o

- GPT-4o (Omni or Universal) <https://openai.com/index/hello-gpt-4o/>
- Multimodal, Multi-lingual Generative Pre-trained Transformer Model
- Released in May 2024

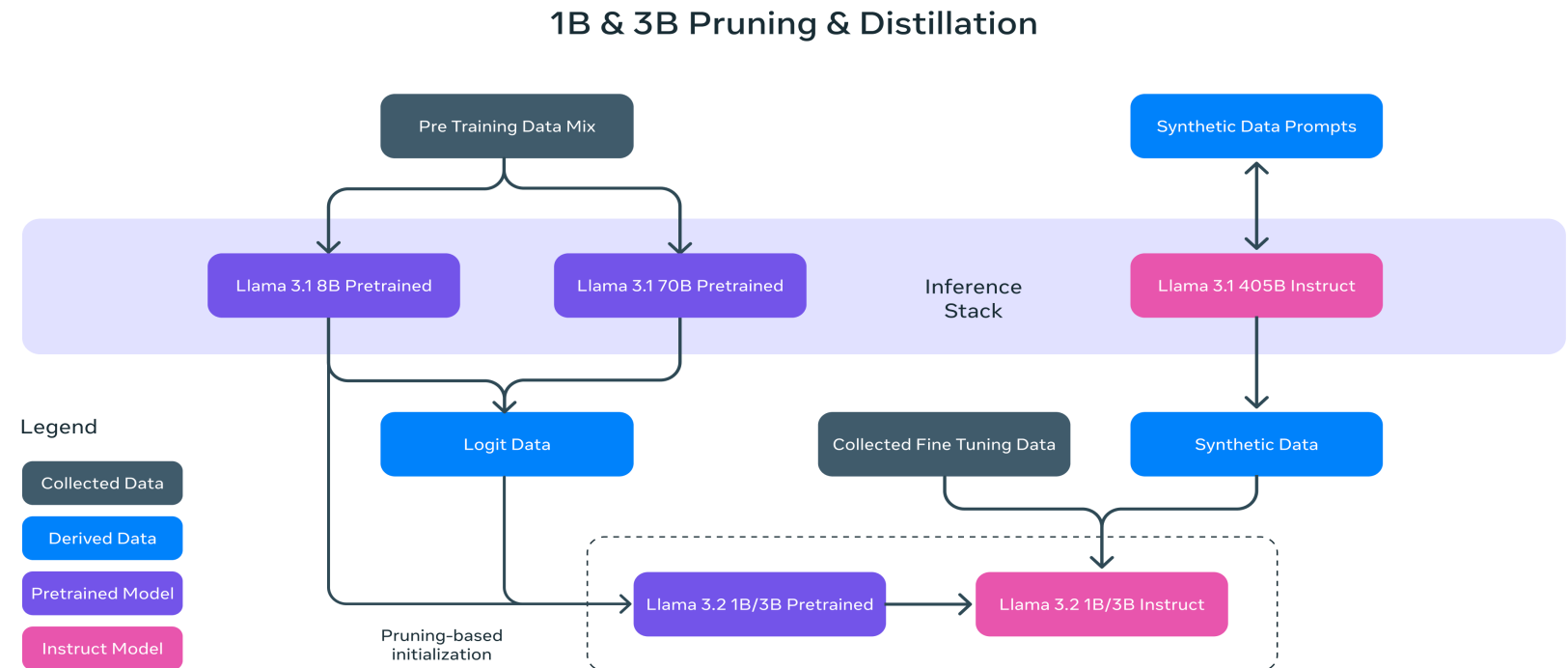


Llama 3.2 <https://www.llama.com>

- Meta's open-source AI model you can fine-tune, distill, and deploy.
- Llama 3.2 is a collection of **large language models (LLMs)** pretrained and fine-tuned in 1B and 3B sizes that are multilingual text only, and 11B and 90B sizes that take both text and image inputs and output text.
- Evaluated performance on over 150 benchmark datasets that span a wide range of languages. For the vision LLMs, evaluated performance on benchmarks for image understanding and visual reasoning.
- Released September, 2024
- Llama 3.2 1B and 90B models support image reasoning use cases, such as document-level understanding including charts and graphs, captioning of images, and visual grounding tasks such as directionally pinpointing objects in images based on natural language descriptions.

Llama 3.2 <https://www.llama.com>

- Key difference between the predecessors' models:
 - size of the pretraining corpus increased by 650% (LLaMA2 was trained on 2T tokens where as LLaMA3 trained on 15T tokens)
 - doubled the context length of the model from 4K to 8K on both 8B and 70B models,
 - adopted grouped-query attention (GQA) for both 8B and 70B variant as compared to the previous generations.



Gemini <https://gemini.google.com/?hl=en-AU>

- Google's largest and most capable AI model
- Natively multimodal - it can generalize and seamlessly understand, operate across and combine different types of information including text, code, audio, image and video.
- With a score of 90.0%, Gemini Ultra is the first model to outperform human experts on MMLU(massive multitask language understanding), which uses a combination of 57 subjects such as math, physics, history, law, medicine and ethics for testing both world knowledge and problem-solving abilities.



Gemini's performance

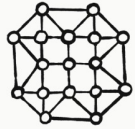
TEXT				
Capability	Benchmark Higher is better	Description	Gemini Ultra	GPT-4 API numbers calculated where reported numbers were missing
General	MMLU	Representation of questions in 57 subjects (incl. STEM, humanities, and others)	90.0% CoT@32*	86.4% 5-shot** (reported)
Reasoning	Big-Bench Hard	Diverse set of challenging tasks requiring multi-step reasoning	83.6% 3-shot	83.1% 3-shot (API)
	DROP	Reading comprehension (F1 Score)	82.4 Variable shots	80.9 3-shot (reported)
	HellaSwag	Commonsense reasoning for everyday tasks	87.8% 10-shot*	95.3% 10-shot* (reported)
Math	GSM8K	Basic arithmetic manipulations (incl. Grade School math problems)	94.4% maj1@32	92.0% 5-shot CoT (reported)
	MATH	Challenging math problems (incl. algebra, geometry, pre-calculus, and others)	53.2% 4-shot	52.9% 4-shot (API)
Code	HumanEval	Python code generation	74.4% 0-shot (IT)*	67.0% 0-shot* (reported)
	Natural2Code	Python code generation. New held out dataset HumanEval-like, not leaked on the web	74.9% 0-shot	73.9% 0-shot (API)
* See the technical report for details on performance with other methodologies				
** GPT-4 scores 87.29% with CoT@32 - see the technical report for full comparison				

MULTIMODAL

Capability	Benchmark	Description Higher is better unless otherwise noted	Gemini	GPT-4V Previous SOTA model listed when capability is not supported in GPT-4V
Image	MMMU	Multi-discipline college-level reasoning problems	59.4% 0-shot pass@1 Gemini Ultra (pixel only*)	56.8% 0-shot pass@1 GPT-4V
	VQAv2	Natural image understanding	77.8% 0-shot Gemini Ultra (pixel only*)	77.2% 0-shot GPT-4V
	TextVQA	OCR on natural images	82.3% 0-shot Gemini Ultra (pixel only*)	78.0% 0-shot GPT-4V
	DocVQA	Document understanding	90.9% 0-shot Gemini Ultra (pixel only*)	88.4% 0-shot GPT-4V (pixel only)
	Infographic VQA	Infographic understanding	80.3% 0-shot Gemini Ultra (pixel only*)	75.1% 0-shot GPT-4V (pixel only)
Video	MathVista	Mathematical reasoning in visual contexts	53.0% 0-shot Gemini Ultra (pixel only*)	49.9% 0-shot GPT-4V
	VATEX	English video captioning (CIDEr)	62.7 4-shot Gemini Ultra	56.0 4-shot DeepMind Flamingo
	Perception Test MCQA	Video question answering	54.7% 0-shot Gemini Ultra	46.3% 0-shot SeVILA
Audio	CoVoST 2 (21 languages)	Automatic speech translation (BLEU score)	40.1 Gemini Pro	29.1 Whisper v2
	FLEURS (62 languages)	Automatic speech recognition (based on word error rate, lower is better)	7.6% Gemini Pro	17.6% Whisper v3

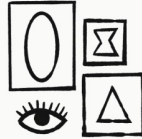
Anthropic Claude 3.5 <https://www.anthropic.com/claude>

Claude's capabilities



Advanced reasoning

Claude can perform complex cognitive tasks that go beyond simple pattern recognition or text generation



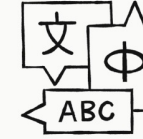
Vision analysis

Transcribe and analyze almost any static image, from handwritten notes and graphs, to photographs



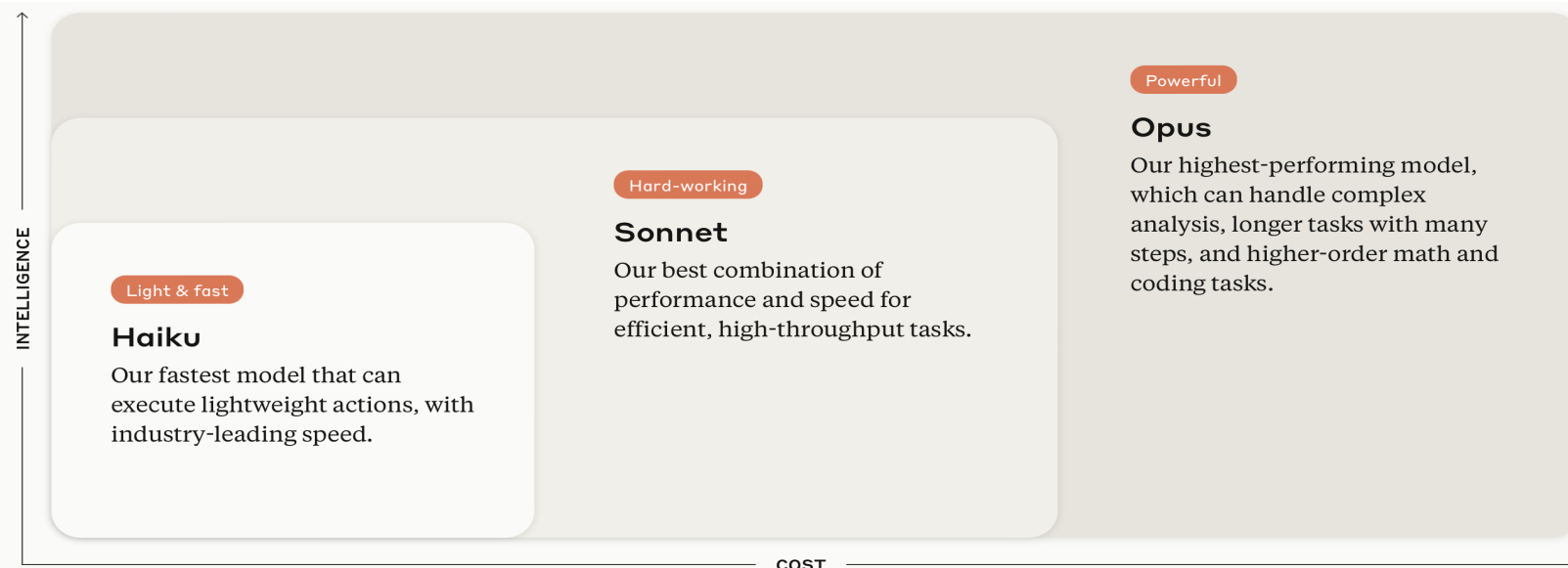
Code generation

Start creating websites in HTML and CSS, turning images into structured JSON data, or debugging complex code bases



Multilingual processing

Translate between various languages in real-time, practice grammar, or create multi-lingual content



Mistral AI <https://mistral.ai/technology/#models>

➤ Build with open-weight models

Mistral Small 24.09

Enterprise-grade small model.

- The most powerful model in its size
- Available under the Mistral Research License
- 128k token context window
- Cost-efficient and fast model for a wide array of use cases such as translation, summarization, and sentiment analysis

Mistral Large 2

Top-tier reasoning for high-complexity tasks, for your most sophisticated needs.

- Multi-lingual (incl. European languages, Chinese, Japanese, Korean, Hindi, Arabic)
- Large context window of 128K tokens
- Native function calling capacities and JSON outputs
- High coding proficiency (80+ coding languages)

Codestral

State-of-the-art Mistral model trained specifically for code tasks.

- Trained on 80+ programming languages (incl. Python, Java, C, C++, PHP, Bash)
- Optimized for low latency: Way smaller than competitive coding models
- Context window of 32K tokens

Mistral Embed

State-of-the-art semantic for extracting representation of text extracts.

- English only for now
- Achieves a retrieval score of 55.26 on the Massive Text Embedding Benchmark (MTEB)

Ministral 3B 24.10

Our most efficient edge model.

The most capable in its category, ideal for low-power, low-latency on-device computing and edge use cases.

- 128k token context window
- Highly capable in function-calling for agentic workflows
- Ideal for lossless quantization and tailoring specific use cases

Ministral 8B 24.10

Our most powerful edge model. Successor to Mistral 7B.

Sets the benchmark in commonsense, reasoning, and function-calling in the sub-10B category.

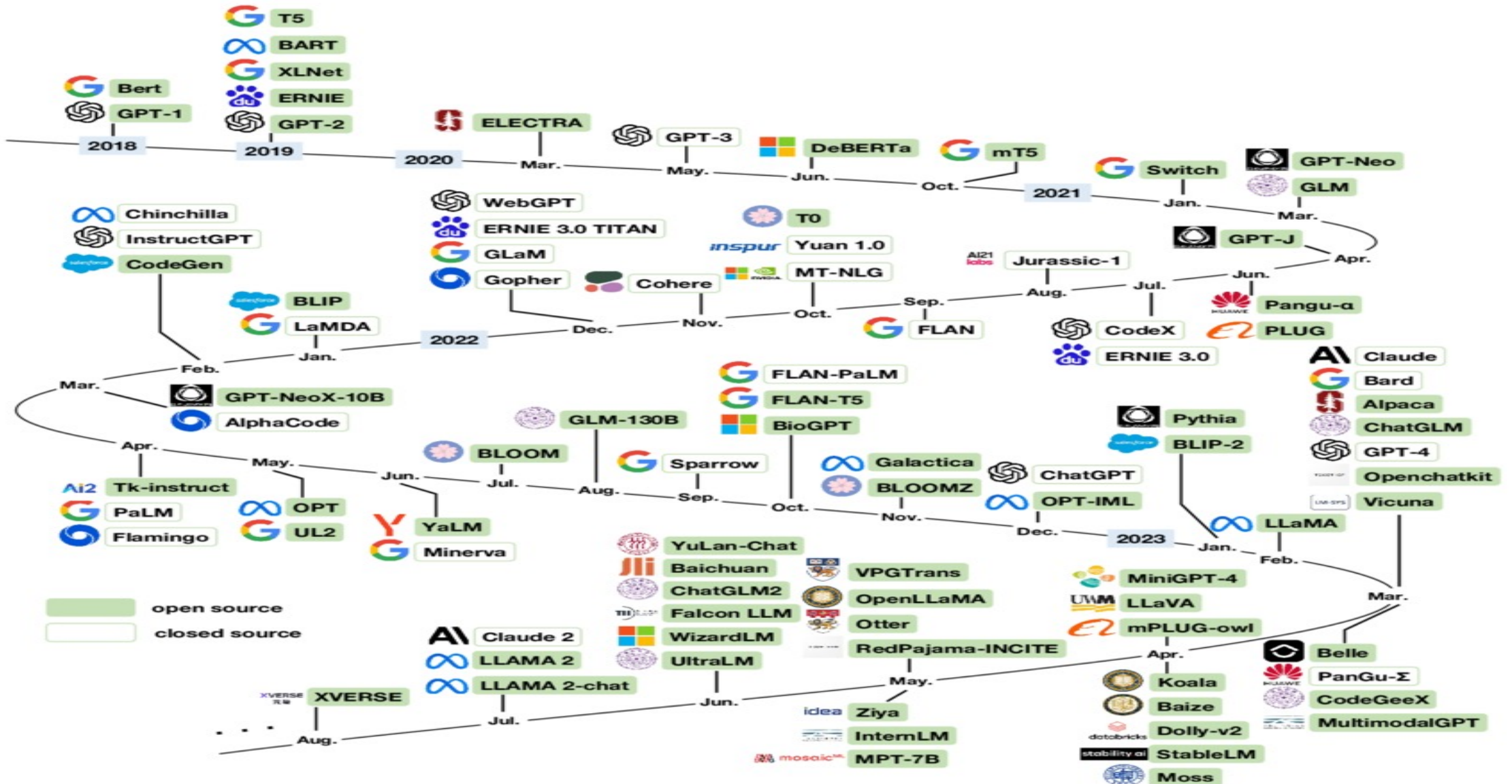
- 128k token context window
- Ideal for on-device computing and edge use cases
- Highly capable in reasoning, function-calling, and common sense

Ernie Bot

- Baidu's Knowledge-Enhanced Large Language Model
- ERNIE Bot was developed based on the ERNIE (Enhanced Representation through Knowledge Integration) and PLATO (Pre-trained Dialogue Generation Model) series models.
- key technologies include supervised fine-tuning, reinforcement learning with human feedback, prompt learning, knowledge enhancement, retrieval enhancement, and dialogue enhancement.



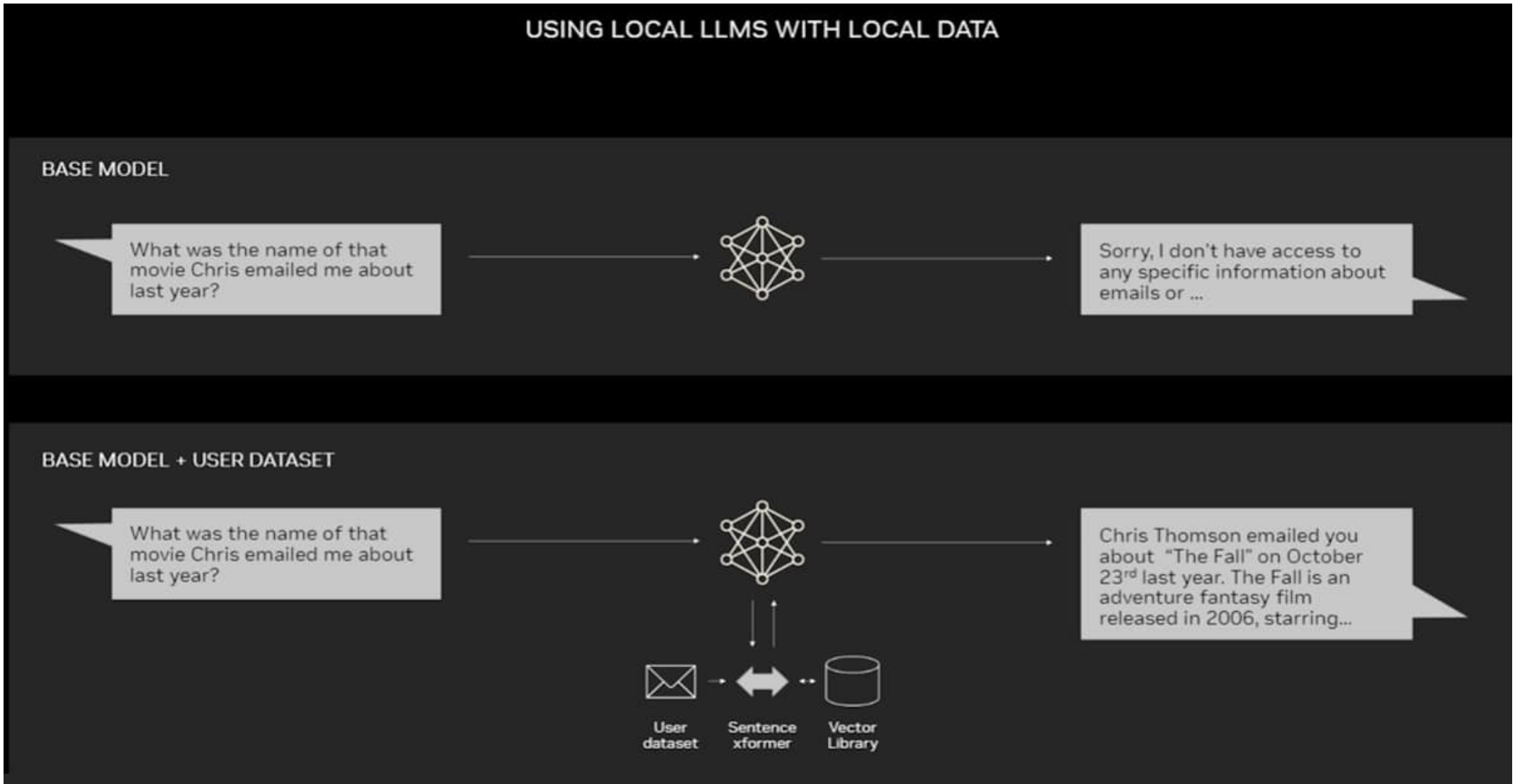
LLMs over the last few years ...



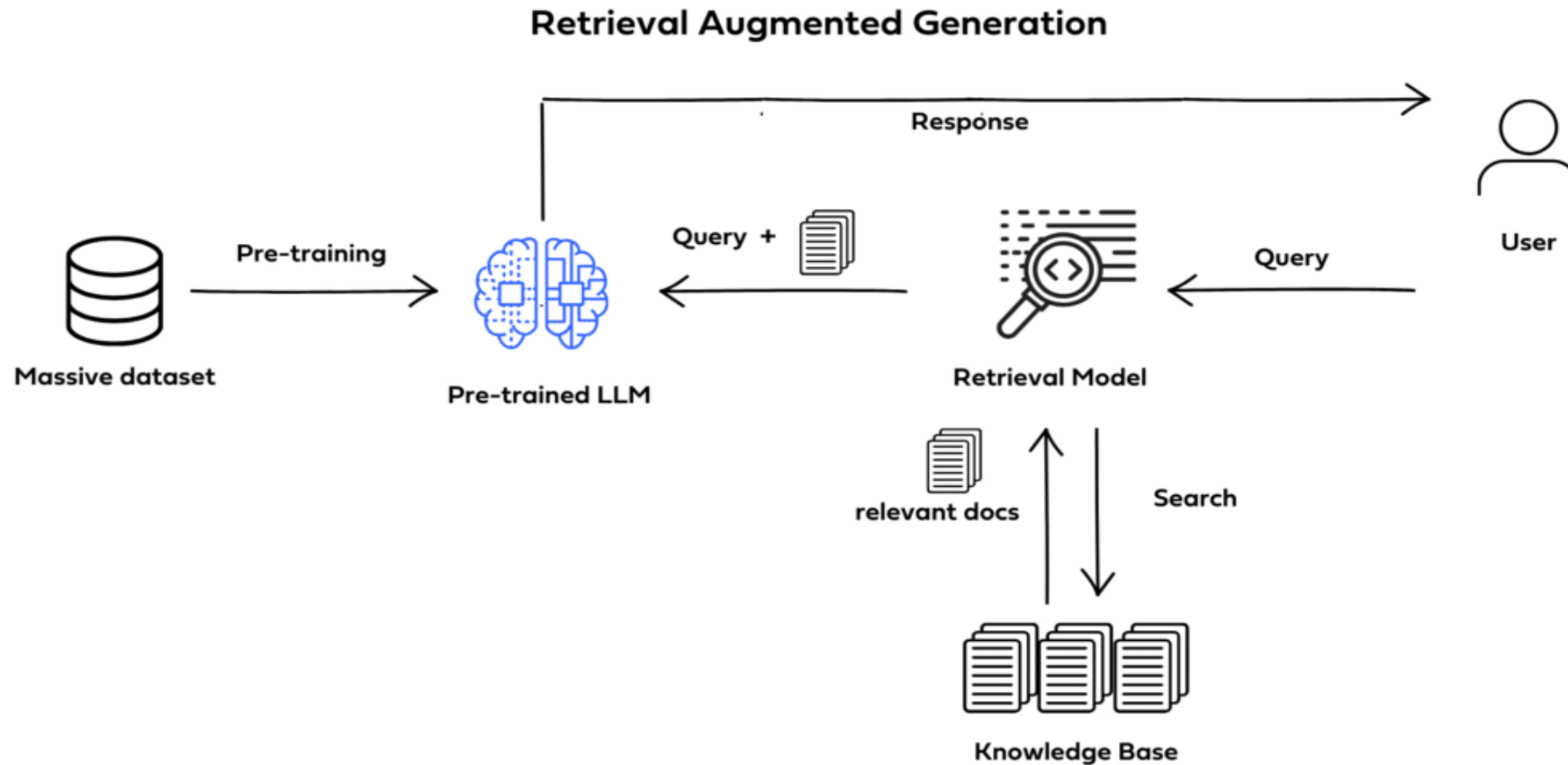
Challenges of LLMs

- Presenting false information when it does not have the answer.
- Presenting out-of-date or generic information when the user expects a specific, current response.
- Creating a response from non-authoritative sources.
- Creating inaccurate responses due to terminology confusion, wherein different training sources use the same terminology to talk about different things.
- Less control over the content
- Data Privacy and Security

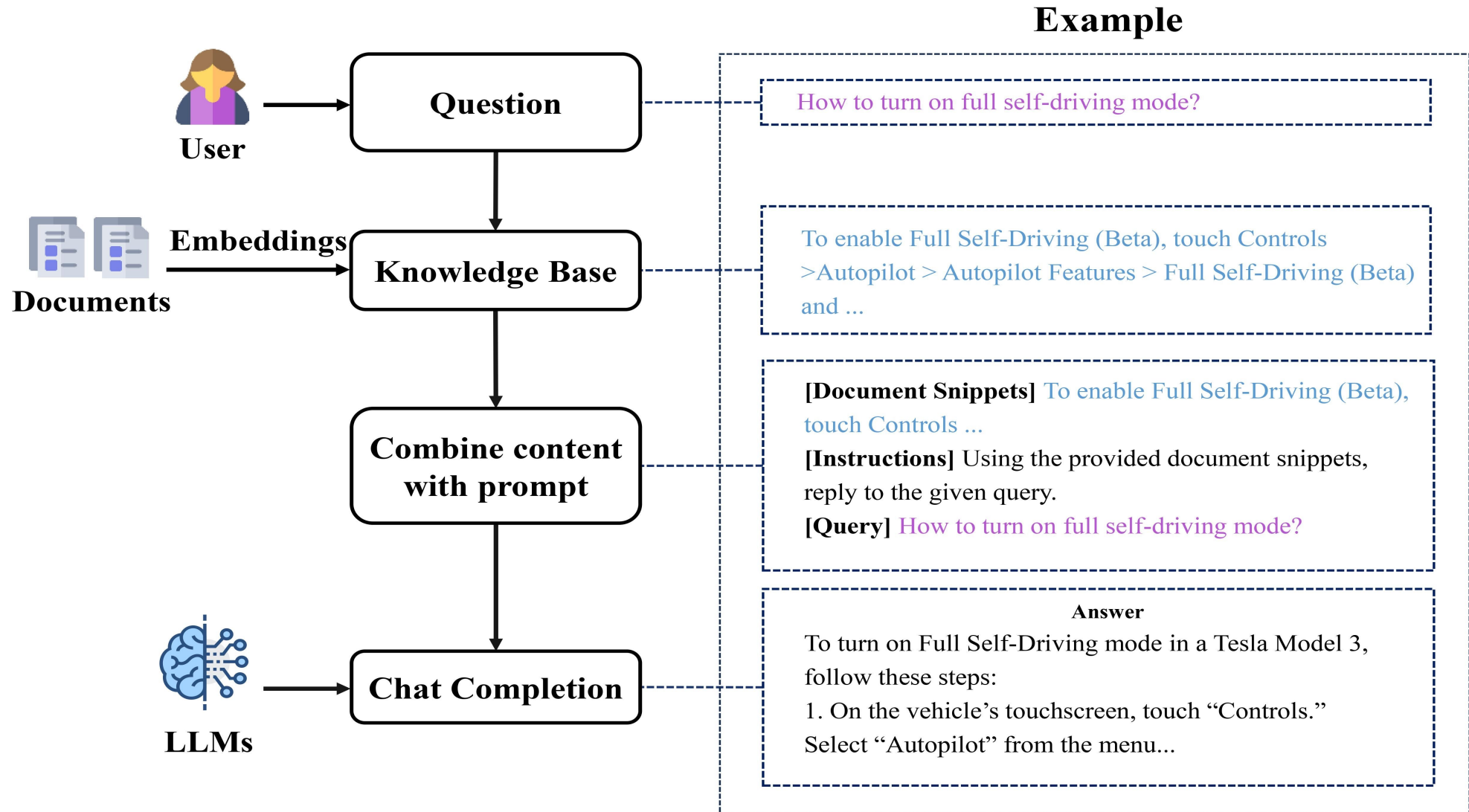
Challenges of LLMs



Retrieval Augmented Generation (RAG)



Retrieval Augmented Generation (RAG)



Some criticism

- Are bigger language models better?
 - Bigger models are better at generalizing on various downstream tasks
 - Issues: Bias, carbon footprints

ARTICLE [OPEN ACCESS](#)

On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? 🦜

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Authors:  [Emily M. Bender](#),  [Timnit Gebru](#),  [Angelina McMillan-Major](#),  [Shmargaret Shmitchell](#) [Authors Info & Claims](#)

FACCT '21: Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency • March 2021 • Pages 610–623 • <https://doi.org/10.1145/3442188.3445922>

Published: 03 March 2021



Yann LeCun

27 October 2020 at 13:01 · 🌐

“It’s entertaining, and perhaps mildly useful as a creative help,” LeCun wrote.

“But trying to build intelligent machines by scaling up language models is like building high-altitude airplanes to go to the moon. You might beat altitude records, but going to the moon will require a completely different approach.”

Large computer language models carry environmental, social risks

It takes an enormous amount of computing power to fuel the model language programs, Bender said. That takes up energy at tremendous scale, and that, the authors argue, causes environmental degradation. And those costs aren’t borne by the computer engineers, but rather by marginalized people who cannot afford the environmental costs.

Another risk comes from the training data itself, the authors say. Because the computers read language from the Web and from other sources, they can pick up and perpetuate racist, sexist, ableist, extremist and other harmful ideologies.

Deep Learning, Deep Pockets?

As you would expect, training a 530-billion parameter model on humongous text datasets requires a fair bit of infrastructure. In fact, Microsoft and NVIDIA used hundreds of DGX A100 multi-GPU servers. At \$199,000 a piece, and factoring in networking equipment, hosting costs, etc., anyone looking to replicate this experiment would have to spend close to \$100 million dollars. Want fries with that?

Summary

- Large language modeling is like race between titans
- Large language model size is increasing 10x every year for the last few years. Moore's Law
- Issues:
 - Environmental cost (carbon footprint)
 - Hard to replicate
 - Very complex
- The need:
 - To build practical and efficient solutions
 - Within everyone's capability and reach to solve real-world problems

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Questions?